

Summary

- Although only a small percentage of all bacteria are responsible for human illness, the suffering and death they cause are significant. Each year, millions of people are infected by pathogenic bacteria using the four major modes of transmission: airborne, arthropod-borne, direct contact, and food-borne and waterborne.
- As the fields of microbiology, immunology, pathology, pharmacology, and epidemiology have expanded current understanding of the disease process, the incidence of many human illnesses has decreased. Many bacterial infections, once the leading cause of death, have been successfully brought under control in most developed countries. Alternatively, several are increasing in incidence throughout the world.
- The bacteria emphasized in this chapter and the diseases they cause are as follows:
 - Airborne diseases
 - diphtheria (*Corynebacterium diphtheriae*) (figure 39.1)
 - Legionnaires' disease and Pontiac fever (*Legionella pneumophila*)
 - meningitis (*Haemophilus influenzae* type b, *Neisseria meningitidis*, and *Streptococcus pneumoniae*)
 - M. avium*–*M. intracellulare* pneumonia
 - Streptococcal diseases (*Streptococcus pyogenes*) (figure 39.3)
 - tuberculosis (*Mycobacterium tuberculosis*) (figure 39.7)
 - pertussis (*Bordetella pertussis*)
 - Arthropod-borne diseases
 - ehrlichiosis (*Ehrlichia chaffeensis*)
 - epidemic (louse-borne) typhus (*Rickettsia prowazekii*)
 - endemic (murine) typhus (*Rickettsia typhi*)
 - Lyme disease (*Borrelia burgdorferi*)
 - plague (*Yersinia pestis*) (figure 39.9)
 - Q fever (*Coxiella burnetii*)
 - Rocky Mountain spotted fever (*Rickettsia rickettsii*)
 - Direct contact diseases
 - anthrax (*Bacillus anthracis*)
 - bacterial vaginosis (*Gardnerella vaginalis*)
 - cat-scratch disease (*Bartonella henselae*)
 - chancroid (*Haemophilus ducreyi*)
 - chlamydial pneumonia (*Chlamydia pneumoniae*)
 - gas gangrene or clostridial myonecrosis (*Clostridium perfringens*)
 - genitourinary mycoplasmal diseases (*Ureaplasma urealyticum*, *Mycoplasma hominis*)
 - gonorrhea (*Neisseria gonorrhoeae*)
 - inclusion conjunctivitis (*Chlamydia trachomatis*)
 - leprosy (*Mycobacterium leprae*)
 - lymphogranuloma venereum (*Chlamydia trachomatis*)
 - mycoplasmal pneumonia (*Mycoplasma pneumoniae*)
 - nongonococcal urethritis (various microorganisms)
 - peptic ulcer disease (*Helicobacter pylori*)
 - psittacosis (ornithosis) (*Chlamydia psittaci*)
 - staphylococcal diseases (*Staphylococcus aureus*) (figure 39.18)
 - syphilis (*Treponema pallidum*)
 - tetanus (*Clostridium tetani*)
 - trachoma (*Chlamydia trachomatis*)
 - tularemia (*Francisella tularensis*)
 - Food-borne and waterborne diseases
 - cholera (*Vibrio cholerae*)
 - botulism (*Clostridium botulinum*)
 - gastroenteritis (*Campylobacter jejuni* and other bacteria)
 - salmonellosis (*Salmonella* serovar typhimurium)
 - shigellosis (*Shigella* spp.)
 - staphylococcal food poisoning (*Staphylococcus aureus*)
 - traveler's diarrhea (*Escherichia coli*)
 - typhoid fever (*Salmonella typhi*)
- Gram-positive bacteria, fungi, and endotoxin-containing gram-negative bacteria can initiate the pathogenic cascade of sepsis leading to septic shock (figure 39.24). Gram-negative sepsis is most commonly caused by *E. coli*, followed by *Klebsiella* spp., *Enterobacter* spp., and *Pseudomonas aeruginosa*.
- Dental plaque formation begins on a tooth with the initial colonization of the acquired enamel pellicle by *Streptococcus gordonii*, *S. oralis*, and *S. mitis*. Other bacteria then become attached and form a plaque ecosystem (figure 39.25). The bacteria produce acids that cause a chemical lesion on the tooth and initiate dental decay or caries. Periodontal disease is a group of diverse clinical entities that affect the periodontium. Disease is initiated by the formation of subgingival plaque, which leads to tissue inflammation known as periodontitis and to periodontal pockets. Bacteria that colonize these pockets can cause an abscess, periodontosis, gingivitis, and general tissue necrosis.

Key Terms

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|-----------------------------------|--|--|
| acquired enamel pellicle 934 | cholera 930 | enteroinvasive <i>E. coli</i> (EIEC) 932 |
| anthrax 913 | choleragen 930 | enteropathogenic <i>E. coli</i> (EPEC) 932 |
| aseptic meningitis syndrome 902 | clostridial myonecrosis 915 | enterotoxigenic <i>E. coli</i> (ETEC) 932 |
| bacterial (septic) meningitis 902 | clue cells 914 | enterotoxin 927 |
| bacterial vaginosis 914 | coaggregation 934 | epidemic (louse-borne) typhus 909 |
| biofilm 920 | congenital syphilis 923 | erysipelas 903 |
| botulism 929 | cutaneous anthrax 913 | eschar 914 |
| Bright's disease 905 | cutaneous diphtheria 901 | exfoliative toxin (exfoliatin) 922 |
| buboes 911 | dental plaque 934 | food-borne infection 926 |
| bubonic plague 911 | diffusely adhering <i>E. coli</i> (DAEC) 932 | food intoxication 927 |
| campylobacteriosis 929 | diphtheria 900 | food poisoning 926 |
| caries 936 | DPT vaccine 901 | gas gangrene 915 |
| caseous lesion 908 | effacing lesions 932 | gastritis 918 |
| cat-scratch disease (CSD) 914 | ehrlichiosis 909 | gastroenteritis 929 |
| cellulitis 903 | endemic (murine) typhus 909 | gastrointestinal anthrax 913 |
| chancre 923 | endogenous infection 905 | genital ulcer disease 914 |
| chancroid 914 | enteroaggregative <i>E. coli</i> (EAaggEC) 932 | Ghon complex 908 |
| chlamydial pneumonia 914 | enterohemorrhagic <i>E. coli</i> (EHEC) 932 | gingivitis 936 |

938 Chapter 39 Human Diseases Caused by Bacteria

glomerulonephritis 905
glucans 936
gonococci 915
gonorrhea 915
gummas 924
hemolytic uremic syndrome 932
impetigo 903
inclusion conjunctivitis 916
Legionnaires' disease (legionellosis) 901
leprous (progressive) leprosy 916
leprosy (Hansen's disease) 916
listeriosis 931
Lyme disease (LD, Lyme borreliosis) 910
lymphogranuloma venereum (LGV) 917
meningitis 902
miliary tuberculosis 908
multi-drug-resistant strains of tuberculosis (MDR-TB) 908
mycoplasmal pneumonia 917
myositis 904
necrotizing fasciitis 904
nongonococcal urethritis (NGU) 918
odontopathogens 933
ophthalmia neonatorum (conjunctivitis of the newborn) 916

pannus 926
pelvic inflammatory disease (PID) 915
peptic ulcer disease 918
periodontal disease 936
periodontitis 936
periodontium 936
periodontosis 936
pertussis 903
pharyngitis 905
plague 911
pneumonic plague 911
Pontiac fever 902
psittacosis (ornithosis) 919
pulmonary anthrax 913
Q fever 912
reactivation tuberculosis 908
rheumatic fever 905
Rocky Mountain spotted fever 913
salmonellosis 931
scarlet fever (scarlatina) 905
sepsis 933
septic shock 933
shigellosis 931
slime 919

staphylococcal food poisoning 932
staphylococcal scalded skin syndrome (SSSS) 922
streptococcal pneumonia 905
streptococcal sore throat 905
subgingival plaque 936
tetanospasmin 924
tetanus 924
tonsillitis 905
toxic shock-like syndrome (TSLs) 904
toxic shock syndrome (TSS) 922
trachoma 925
transovarian passage 913
traveler's diarrhea 932
tubercles 908
tuberculoid (neural) leprosy 916
tuberculosis (TB) 906
tuberculous cavities 908
tularemia 926
typhoid fever 933
vasculitis 909
venereal syphilis 923
Weil-Felix reaction 910

Questions for Thought and Review

- Briefly describe each of the major or most common bacterial diseases in terms of its causative agent, signs and symptoms, the course of infection, mechanism of pathogenesis, epidemiology, and prevention and/or treatment.
- Because there are many etiologies of gastroenteritis, how is a definitive diagnosis usually made?
- Differentiate between the following factors of bacterial intoxication and infection: etiologic agents, onset, duration, symptoms, and treatment.
- How would you differentiate between salmonellosis and botulism?
- Why do urinary tract infections frequently occur after catheterization procedures?
- Why is botulism the most serious form of bacterial food poisoning?
- Why is tuberculosis still a problem in underdeveloped countries? In the United States?
- What is the standard treatment for most bacterial diarrheas?
- Why are most cases of gastroenteritis not treated with antibiotics?
- How are tetanus, gas gangrene, and botulism related?
- Give several reasons why treatment of tuberculosis is so difficult.
- Describe the various bacteria that cause food poisoning and give each one's most probable source.
- Explain why Lyme disease is receiving increased attention in the United States.
- Why is septic shock caused by gram-negative sepsis increasing in prevalence in the United States?
- What is the major difference between trachoma and inclusion conjunctivitis?
- Why is Rocky Mountain spotted fever the most important rickettsial disease in the United States?
- As a group of diseases, what are some of the common characteristic symptoms and pathological results of rickettsial disease?
- Why is it difficult to definitely isolate the microorganism responsible for nongonococcal urethritis?
- Why are dental diseases relatively new infectious diseases of humans?
- What makes Q fever a unique rickettsial disease?
- What specific chlamydial and mycoplasmal diseases are transmitted sexually?

Critical Thinking Questions

- Why is tetanus a concern only when one has a deep puncture-type wound and not a surface cut or abrasion?
- Think about our modern, Western lifestyles. Can you name and describe bacterial diseases that result from this life of relative luxury? Refer to *Infections of Leisure*, second edition, edited by David Schlossberg (1999), published by the American Society for Microbiology Press.
- You have been assigned the task of eradicating gonorrhea in your community. Explain how you would accomplish this.
- You are a park employee. How would you prevent people from acquiring arthropod-borne diseases?

Additional Reading

General

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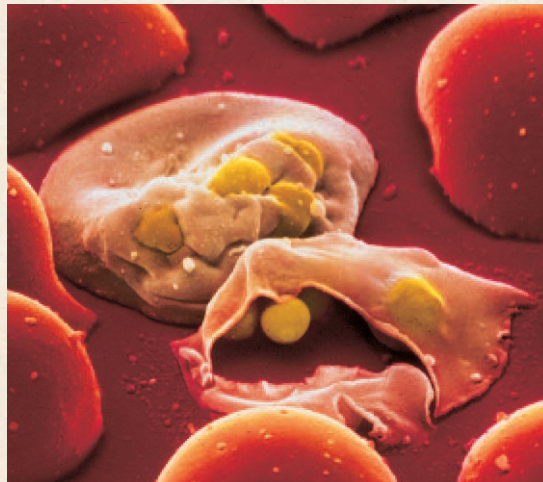
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CHAPTER 40

Human Diseases Caused by Fungi and Protozoa



This photograph shows malaria parasites (yellow) bursting out of red blood cells. Malaria is one of the worst scourges of humanity. Indeed, malaria has played an important part in the rise and fall of nations (see the chapter opening quote), and has killed untold millions the world over. Despite the combined efforts of 102 countries to eradicate malaria, it remains the most important disease in the world today in terms of lives lost and economic burden.

Outline

- 40.1 Fungal Diseases 942
 - Superficial Mycoses 942
 - Cutaneous Mycoses 943
 - Subcutaneous Mycoses 945
 - Systemic Mycoses 945
 - Opportunistic Mycoses 948
- 40.2 Protozoan Diseases 950
 - Amebiasis 950
 - Cryptosporidiosis 952
 - Freshwater Amoeba Diseases 953
 - Giardiasis 953
 - Malaria 954
 - Hemoflagellate Diseases 956
 - Toxoplasmosis 957
 - Trichomoniasis 958

Concepts

1. Fungal diseases (mycoses) are usually divided into five groups according to the level of infected tissue and mode of entry into the host: (1) superficial, (2) cutaneous, (3) subcutaneous, (4) systemic, and (5) opportunistic infections.
2. The superficial mycoses occur mainly in the tropics and include black piedra, white piedra, and tinea versicolor.
3. The cutaneous mycoses—that is, those of the outer layer of the skin—are generally called ringworms, tineas, or dermatomycoses. These diseases occur worldwide and represent the most common fungal diseases in humans.
4. The dermatophytes that cause the subcutaneous—below the skin—mycoses are normal saprophytic inhabitants of the soil. They must be introduced into the body beneath the cutaneous layer. Examples of these diseases include chromomycosis, maduromycosis, and sporotrichosis.
5. The systemic mycoses are the most serious of the fungal infections in the normal host because the responsible fungi can disseminate throughout the body. Examples include blastomycosis, coccidioidomycosis, cryptococcosis, and histoplasmosis.
6. The opportunistic mycoses can create life-threatening situations in the compromised host. Examples of these diseases include aspergillosis, candidiasis, and *Pneumocystis carinii* pneumonia.

Concepts *(Continued)*

7. About 20 different protozoa cause human diseases that afflict hundreds of millions of people throughout the world. Examples include amebiasis, cryptosporidiosis, giardiasis, malaria, the hemoflagellate diseases, toxoplasmosis, and trichomoniasis.
8. Certain fungal and protozoan diseases are increasing in incidence because of organ transplants, immunosuppressive drugs, and AIDS.

Historians believe that malaria has probably had a greater impact on world history than any other infectious disease, influencing the outcome of wars, various population movements, and the development and decline of various civilizations.

—Lynne S. Garcia

The purpose of this chapter is to describe some of the fungal and protozoan microorganisms that are pathogenic to humans and to discuss the clinical manifestations, diagnosis, epidemiology, pathogenesis, and treatment of the diseases caused by them.

Besides the viruses and bacteria, two other major groups of microorganisms cause infectious diseases in humans: fungi and protozoa. The biology of these organisms is covered in chapters 25 and 27, respectively. This chapter emphasizes the diseases they cause.

40.1 Fungal Diseases

Although hundreds of thousands of fungal species are found in the environment, only about 50 can produce disease in humans. **Medical mycology** is the discipline that deals with the fungi that causes human disease. These fungal diseases, known as **mycoses** [s., mycosis; Greek *mykes*, fungus], are divided into five groups according to the type of infected tissue in the host: superficial, cutaneous, subcutaneous, systemic, and opportunistic mycoses (tables 40.1 and 40.2). It should be noted that the importance of opportunistic fungal pathogens is increasing because of the expansion of the immunocompromised patient population.

Superficial Mycoses

The superficial mycoses are extremely rare in the United States, and most occur in the tropics. The fungi responsible are limited to the outer surface of hair and skin and hence are called superficial. Infections of the hair shaft are collectively

Table 40.1 Examples of Some Medically Important Fungi

Group	Pathogen	Location	Disease
Superficial mycoses	<i>Piedraia hortae</i>	Scalp	Black piedra
	<i>Trichosporon beigelii</i>	Beard, mustache	White piedra
	<i>Malassezia furfur</i>	Trunk, neck, face, arms	Tinea versicolor
Cutaneous mycoses	<i>Trichophyton mentagrophytes</i> , <i>T. verrucosum</i> , <i>T. rubrum</i>	Beard hair	Tinea barbae
	<i>Trichophyton</i> , <i>Microsporum canis</i>	Scalp hair	Tinea capitis
	<i>Trichophyton rubrum</i> , <i>T. mentagrophytes</i> , <i>Microsporum canis</i>	Smooth or bare parts of the skin	Tinea corporis
	<i>Epidermophyton floccosum</i> , <i>T. mentagrophytes</i> , <i>T. rubrum</i>	Groin, buttocks	Tinea cruris (jock itch)
	<i>T. rubrum</i> , <i>T. mentagrophytes</i> , <i>E. floccosum</i>	Feet	Tinea pedis (athlete's foot)
	<i>T. rubrum</i> , <i>T. mentagrophytes</i> , <i>E. floccosum</i>	Nails	Tinea unguium (onychomycosis)
Subcutaneous mycoses	<i>Phialophora verrucosa</i> , <i>Fonsecaea pedrosoi</i>	Legs, feet	Chromoblastomycosis
	<i>Madurella mycetomatis</i>	Feet, other areas of body	Maduromycosis
	<i>Sporothrix schenckii</i>	Puncture wounds	Sporotrichosis
Systemic mycoses	<i>Blastomyces dermatitidis</i>	Lungs, skin	Blastomycosis
	<i>Coccidioides immitis</i>	Lungs, other parts of body	Coccidioidomycosis
	<i>Cryptococcus neoformans</i>	Lungs, skin, bones, viscera, central nervous system	Cryptococcosis
	<i>Histoplasma capsulatum</i>	Within phagocytes	Histoplasmosis
Opportunistic mycoses	<i>Aspergillus fumigatus</i> , <i>A. flavus</i>	Respiratory system	Aspergillosis
	<i>Candida albicans</i>	Skin or mucous membranes	Candidiasis
	<i>Pneumocystis carinii</i>	Lungs, sometimes brain	Pneumocystis pneumonia

Table 40.2 Examples of Some Human Fungal Diseases Recognized Since 1973

Year	Fungus	Disease
Molds		
1974	<i>Phialophora parasitica</i>	Phaeohyphomycosis
1992	<i>Penicillium marneffei</i>	Disseminated infection
Yeasts		
1989	<i>Candida lusitanae</i>	Fungemia
1989	<i>Malassezia furfur</i>	Fungemia
1990	<i>Rhodotorula rubra</i>	Fungemia
1991	<i>Candida ciferrii</i>	Fungemia
1993	<i>Hansenula anomala</i>	Fungemia
1993	<i>Trichosporon beigeli</i>	Fungemia

called **pedras** (Spanish for stone because they are associated with the hard nodules formed by mycelia [see figure 25.4] on the hair shaft). For example, **black piedra** is caused by *Piedraia hortae* and forms hard black nodules on the hairs of the scalp (**figure 40.1**). **White piedra** is caused by the yeast *Trichosporon beigeli* and forms light-colored nodules on the beard and mustache. Some superficial mycoses are called **tineas** [Latin for grub, larva, worm], the specific type being designated by a modifying term. Tineas are superficial fungal infections involving the outer layers of the skin, nails, and hair. **Tinea versicolor** is caused by the yeast *Malassezia furfur* and forms brownish-red scales on the skin of the trunk, neck, face, and arms. Treatment involves removal of the skin scales with a cleansing agent and removal of the infected hairs. Good personal hygiene prevents these infections.

Cutaneous Mycoses

Cutaneous mycoses—also called **dermatomycoses**, **ringworms**, or **tineas**—occur worldwide and represent the most common fungal diseases in humans. Three genera of cutaneous fungi, or **dermatophytes**, are involved in these mycoses: *Epidermophyton*, *Microsporum*, and *Trichophyton*. Diagnosis is by microscopic examination of biopsied areas of the skin cleared with 10% potassium hydroxide and by culture on Sabouraud dextrose agar. Treatment is with topical ointments such as miconazole (Monistat-Derm), tolnaftate (Tinactin), or clotrimazole (Lotrimin) for 2 to 4 weeks. Griseofulvin (Grifulvin V) and itraconazole (Sporanox) are the only oral fungal agents currently approved by the FDA for treating dermatophytoses. [The mode of action of ketoconazole, miconazole, tolnaftate, and griseofulvin \(p. 820\)](#)

Tinea barbae [Latin *barba*, the beard] is an infection of the beard hair (**figure 40.2**) caused by *Trichophyton mentagrophytes* or *T. verrucosum*. It is predominantly a disease of men who live in rural areas and acquire the fungus from infected animals.

Tinea capitis [Latin *capita*, the head] is an infection of the scalp hair (**figure 40.3a**). It is characterized by loss of hair, inflammation, and scaling. Tinea capitis is primarily a childhood

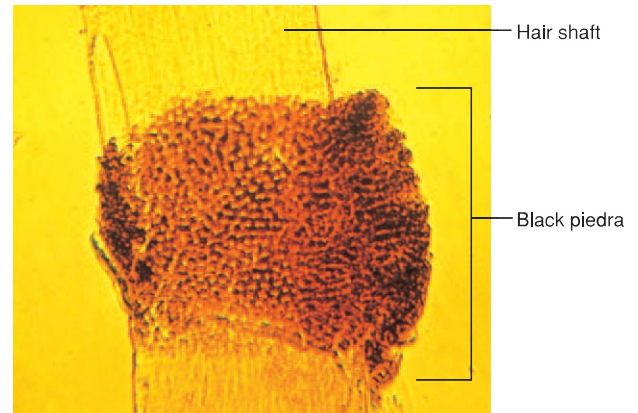


Figure 40.1 Superficial Mycosis: Black Piedra. Hair shaft infected with *Piedraia hortae*; light micrograph ($\times 200$).



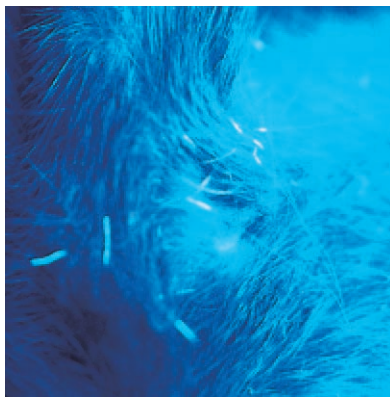
Figure 40.2 Cutaneous Mycosis: Tinea Barbae. Ringworm of the beard caused by *Trichophyton mentagrophytes*.

disease caused by *Trichophyton* or *Microsporum* species. Person-to-person transmission of the fungus occurs frequently when poor hygiene and overcrowded conditions exist. The fungus also occurs in domestic animals, from whom it can be transmitted to humans. A Wood's light (a UV light) can help with the diagnosis of tinea capitis because fungus-infected hair fluoresces when illuminated by UV radiation (**figure 40.3b**).

Tinea corporis [Latin *corpus*, the body] is a dermatophytic infection of the smooth or bare parts of the skin (**figure 40.4**). The disease is characterized by circular, red, well-demarcated, scaly, vesiculopustular lesions accompanied by itching. Tinea corporis is caused by *Trichophyton rubrum*, *T. mentagrophytes*, or *Microsporum canis*.



(a)



(b)

Figure 40.3 Cutaneous Mycosis: Tinea Capitis. (a) Ringworm of the head caused by *Microsporum audouinii*. (b) Close-up utilizing a Wood's light.

Transmission of the disease is by direct contact with infected animals (see figure 25.1b) or humans or by indirect contact through fomites.

Tinea cruris [Latin *crura*, the leg] is a dermatophytic infection of the groin (figure 40.5). The pathogenesis and clinical manifestations are similar to those of tinea corporis. The responsible fungi are *Epidermophyton floccosum*, *T. mentagrophytes*, or *T. rubrum*. Factors predisposing one to recurrent disease are moisture, occlusion, and skin trauma. Wet bathing suits, athletic supporters (**jock itch**), tight-fitting slacks, panty hose, and obesity are frequently contributing factors.

Tinea pedis [Latin *pes*, the foot], also known as **athlete's foot**, and **tinea manuum** [Latin *mannus*, the hand] are dermatophytic infections of the feet (figure 40.6) and hands, respectively. Clinical symptoms vary from a fine scale to a vesiculopustular eruption. Itching is frequently present. Warmth, humidity, trauma, and occlusion increase susceptibility to infection. Most infections are caused by *T. rubrum*, *T. mentagrophytes*, or *E. floccosum*. Tinea



Figure 40.4 Cutaneous Mycosis: Tinea Corporis. Ringworm of the body—in this case, the forearm—caused by *Trichophyton mentagrophytes*. Notice the circular patches (arrows).



Figure 40.5 Cutaneous Mycosis: Tinea Cruris. Ringworm of the groin caused by *Epidermophyton floccosum*.

pedis and tinea manuum occur throughout the world, are most commonly found in adults, and increase in frequency with age.

Tinea unguium [Latin *unguis*, nail] is a dermatophytic infection of the nail bed (figure 40.7). In this disease the nail becomes discolored and then thickens. The nail plate rises and separates from the nail bed. *Trichophyton rubrum* or *T. mentagrophytes* are the causative fungi.



Figure 40.6 Cutaneous Mycosis: Tinea Pedis. Ringworm of the foot caused by *Trichophyton rubrum*, *T. mentagrophytes*, or *Epidermophyton floccosum*.



Figure 40.8 Subcutaneous Mycosis. Chromoblastomycosis of the foot caused by *Fonsecaea pedrosoi*.



Figure 40.7 Cutaneous Mycosis: Tinea Unguium. Ringworm of the nails caused by *Trichophyton rubrum*.

Subcutaneous Mycoses

The dermatophytes that cause subcutaneous mycoses are normal saprophytic inhabitants of soil and decaying vegetation. Because they are unable to penetrate the skin, they must be introduced into the subcutaneous tissue by a puncture wound that has been contaminated with soil containing the fungi. Most infections involve barefooted agricultural workers. Once in the subcutaneous tissue, the disease develops slowly—often over a period of years. During this time the fungi produce a nodule that eventually ulcerates and the organisms spread along lymphatic channels producing more subcutaneous nodules. At times such nodules drain to the skin surface. The administration of oral 5-fluorocytosine, iodides,

amphotericin B, and surgical excision are the usual treatments. Diagnosis is accomplished by culture of the infected tissue. [The mode of action of amphotericin B \(p. 820\)](#)

One type of subcutaneous mycosis is **chromoblastomycosis**. The nodules are pigmented a dark brown. This disease is caused by the black molds *Phialophora verrucosa* or *Fonsecaea pedrosoi*. These fungi exist worldwide, especially in tropical and subtropical regions. Most infections involve the legs and feet (**figure 40.8**).

Another subcutaneous mycosis is **maduromycosis**, caused by *Madurella mycetomatis*, which is distributed worldwide and is especially prevalent in the tropics. Because the fungus destroys subcutaneous tissue and produces serious deformities, the resulting infection is often called a **eumycotic mycetoma** or fungal tumor (**figure 40.9**). One form of mycetoma, known as Madura foot, occurs through skin abrasions acquired while walking barefoot on contaminated soil.

Sporotrichosis is the subcutaneous mycosis caused by the dimorphic (*see table 25.2*) fungus *Sporothrix schenckii*. The disease occurs throughout the world and is the most common subcutaneous mycotic disease in the United States. The fungus can be found in the soil, on living plants, such as barberry shrubs and roses, or in plant debris, such as sphagnum moss and pine-bark mulch. Infection occurs by a puncture wound from a thorn or splinter contaminated with the fungus. The disease is an occupational hazard to florists, gardeners, and forestry workers. After an incubation period of 1 to 12 weeks, a small red papule arises and begins to ulcerate (**figure 40.10**). New lesions appear along lymph channels and can remain localized or spread throughout the body, producing **extracutaneous sporotrichosis**.

Systemic Mycoses

Except for *Cryptococcus neoformans*, which has only a yeast form, the fungi that cause the systemic or deep mycoses are dimorphic—that is, they exhibit a parasitic yeastlike phase (Y) and a saprophytic



Figure 40.9 Subcutaneous Mycosis. Eumycotic mycetoma of the foot caused by *Madurella mycetomatis*.



Figure 40.11 Systemic Mycosis. Blastomycosis of the forearm caused by *Blastomyces dermatitidis*.



Figure 40.10 Subcutaneous Mycosis. Sporotrichosis of the arm caused by *Sporothrix schenckii*.

mold or mycelial phase (M). (See the YM shift, section 25.3.) Most systemic mycoses are acquired by the inhalation of spores from soil in which the mold-phase of the fungus resides. If a susceptible person inhales enough spores, an infection begins as a lung lesion, becomes chronic, and spreads through the bloodstream to other organs (the target organ varies with the species).

Blastomycosis is the systemic mycosis caused by *Blastomyces dermatitidis*, a fungus that grows as a budding yeast in humans but as a mold on culture media and in the environment. It is found predominately in the soil of the Mississippi and Ohio River basins. The disease occurs in three clinical forms: cutaneous, pulmonary, and disseminated. The initial infection begins when blastospores (see figure 25.7f) are inhaled into the lungs. The fungus can then spread rapidly, especially to the skin, where cutaneous ulcers and abscess formation occur (figure 40.11). *B. dermatitidis* can be isolated from pus and biopsy sections. Diagnosis re-

quires the demonstration of thick-walled, yeastlike cells 8 to 15 μm in diameter. Complement-fixation, immunodiffusion, and skin (blastomycin) tests are also useful. Amphotericin B (Fungi-zone), itraconazole (Sporanox), or ketoconazole (Nizoral) are the drugs of choice for treatment. Surgery may be necessary for the drainage of large abscesses. Approximately 30 to 60 deaths are reported each year in the United States. There are no preventive or control measures.

Coccidioidomycosis, also known as valley fever, San Joaquin fever, or desert rheumatism because of the geographical distribution of the fungus, is caused by *Coccidioides immitis*. *C. immitis* exists in the dry, highly alkaline soils of North, Central, and South America. It has been estimated that in the United States about 100,000 people are infected annually with 50 to 100 deaths. Endemic areas have been defined by massive skin testing with the antigen coccidioidin. In the soil and on culture media, this fungus grows as a mold that forms arthroconidia (see figure 25.7b) at the tips of hyphae. Arthroconidia are so abundant in these endemic areas that by simply moving through such an area, one can acquire the disease by inhaling them. Wind turbulence and even construction of outdoor structures have been associated with increased exposure and infection. In humans the fungus grows as a yeast-forming, thick-walled spherule filled with endospores (figure 40.12). Most cases of coccidioidomycosis are asymptomatic or indistinguishable from ordinary upper respiratory infections. Almost all cases resolve themselves in a few weeks, and a lasting immunity results. A few infections result in a progressive chronic pulmonary disease. The fungus also can spread throughout the body, involving almost any organ or site. Diagnosis is accomplished by aspiration and identification of the large spherules (approximately 80 μm in diameter) in pus, sputum, and aspirates. Culturing clinical samples in the presence of penicillin and streptomycin on Sabouraud agar also is diagnostic. Newer methods of rapid confirmation include the testing of supernatants of liquid media cultures for antigens, serology, and skin testing. Miconazole (Lotrimin), itraconazole, ketoconazole, and amphotericin B

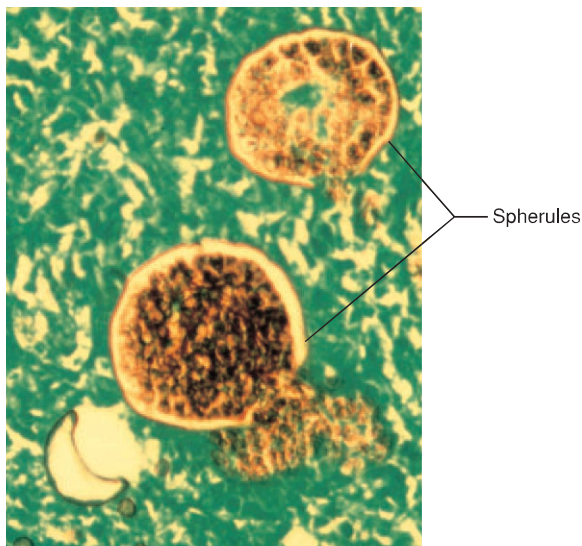


Figure 40.12 Systemic Mycosis: Coccidioidomycosis.

Coccidioides immitis mature spherules filled with endospores within a tissue section; light micrograph ($\times 400$).

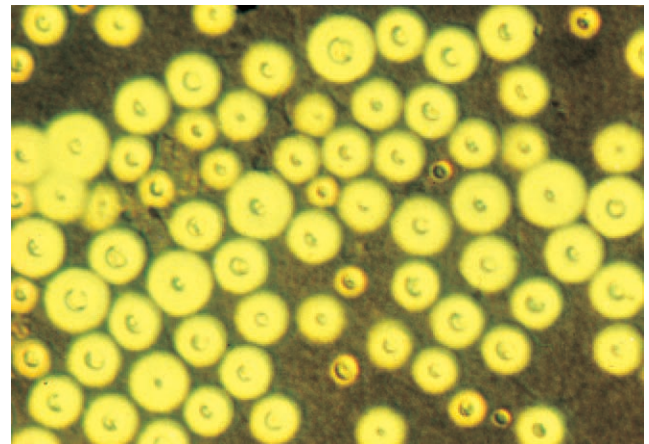


Figure 40.13 Systemic Mycosis: Cryptococcosis. India ink preparation showing *Cryptococcus neoformans*. Although these microorganisms are not budding, they can be differentiated from artifacts by their doubly refractile cell walls, distinctly outlined capsules surrounding all cells, and refractile inclusions in the cytoplasm; light micrograph ($\times 150$).

are the drugs of choice for treatment. Prevention involves reducing exposure to dust (soil) in endemic areas.

Cryptococcosis is a systemic mycosis caused by *Cryptococcus neoformans*. This fungus always grows as a large budding yeast. In the environment *C. neoformans* is a saprophyte with a worldwide distribution. Aged, dried pigeon droppings are an apparent source of infection. Cryptococcosis is found in approximately 15% of AIDS patients. The fungus enters the body by the respiratory tract, causing a minor pulmonary infection that is usually transitory. Some pulmonary infections spread to the skin, bones, viscera, and central nervous system. Once the nervous system is involved, cryptococcal meningitis usually results. Diagnosis is accomplished by detection of the thick-walled spherical yeast cells (**figure 40.13**) in pus, sputum, or exudate smears using India ink to define the organism. The fungus can be easily cultured on Sabouraud dextrose agar. Identification of the fungus in body fluids is made by immunologic procedures. Treatment includes amphotericin B or itraconazole. There are no preventive or control measures.

Histoplasmosis is caused by *Histoplasma capsulatum* var. *capsulatum*, a facultative parasitic fungus that grows intracellularly. It appears as a small budding yeast in humans and on culture media at 37°C. At 25°C it grows as a mold, producing small microconidia (1 to 5 μm in diameter) that are borne singly at the tips of short conidiophores. Large macroconidia or chlamydospores (8 to 16 μm in diameter) are also formed on conidiophores (**figure 40.14a**). In humans the yeastlike form grows within phagocytic cells (**figure 40.14b**). *H. capsulatum* var. *capsulatum* is found as the mycelial form in soils throughout the world and is localized in areas that have been contaminated with bird or bat excrement. The chlamydospores, particularly the mi-

croconidia, are easily spread by air currents. Within the United States, histoplasmosis is endemic within the Mississippi, Kentucky, Tennessee, Ohio, and Rio Grande River basins. More than 75% of the people who reside in parts of these areas have antibodies against the fungus. It has been estimated that in endemic areas of the United States, about 500,000 individuals are infected annually: 50,000 to 200,000 become ill; 3,000 require hospitalization; and about 50 die. The total number of infected individuals may be over 40 million in the United States alone. Histoplasmosis is an occupational disease among spelunkers (people who explore caves) and bat guano miners.

Humans acquire histoplasmosis from airborne microconidia that are produced under favorable environmental conditions. Microconidia are most prevalent where bird droppings, or guanos—especially from starlings, crows, blackbirds, cowbirds, sea gulls, turkeys, and chickens—have accumulated. It is noteworthy that the birds themselves are not infected because of their high body temperature; their droppings simply provide the nutrients for this fungus. Only bats and humans demonstrate the disease and harbor the fungus.

Histoplasmosis is a disease of the monocyte-macrophage system; thus many organs of the body can be infected (*see figure 31.4*). More than 95% of “histo” cases have either no symptoms or mild symptoms such as coughing, fever, and joint pain. Lesions may appear in the lungs and show calcification; most infections resolve on their own. Only rarely does the disease disseminate.

Laboratory diagnosis is accomplished by complement-fixation tests and isolation of the fungus from tissue specimens. Most individuals with this disease exhibit a hypersensitive state that can be demonstrated by the histoplasmin skin test. Currently

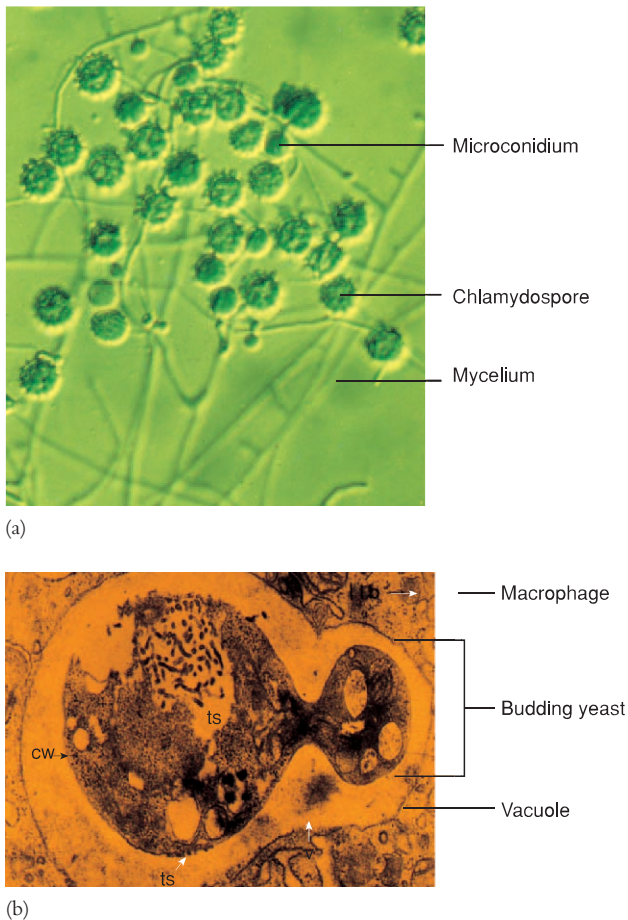


Figure 40.14 Morphology of *Histoplasma capsulatum* var. *capsulatum*. (a) Mycelia, microconidia, and chlamydospores as found in the soil. These are the infectious particles; light micrograph ($\times 125$). (b) Yeastlike cells in a macrophage. Budding *H. capsulatum* within a vacuole. Tubular structures, ts, are observed beneath the cell wall, cw; electron micrograph ($\times 23,000$).

the most effective treatment is with amphotericin B, ketoconazole, or itraconazole. Prevention and control involve wearing protective clothing and masks before entering or working in infested habitats. Soil decontamination with 3 to 5% formalin is effective where economically and physically feasible.

Opportunistic Mycoses

An **opportunistic microorganism** is generally harmless in its normal environment but becomes pathogenic in a compromised host. A **compromised host** is seriously debilitated and has a lowered resistance to infection. There are many causes of this condition, among them the following: malnutrition; alcoholism; cancer, diabetes, leukemia, or another infectious disease; trauma



Figure 40.15 An Opportunistic Mycosis. Aspergillosis of the eye caused by *Aspergillus fumigatus*.

from surgery or injury; an altered microbiota from the prolonged use of antibiotics (e.g., in vaginal candidiasis); and immunosuppression by drugs, viruses (HIV), hormones, genetic deficiencies, chemotherapy of patients, and old age. The most important opportunistic mycoses include systemic aspergillosis, candidiasis, and *Pneumocystis carinii* pneumonia.

Of all the fungi that cause disease in compromised hosts, none are as widely distributed as the *Aspergillus* species. *Aspergillus* is omnipresent in nature, being found wherever organic debris occurs. *Aspergillus fumigatus* is the usual cause of **aspergillosis**. *A. flavus* is the second most important species, particularly in invasive disease of immunosuppressed patients.

The major portal of entry for *Aspergillus* is the respiratory tract. Inhalation of conidiospores (see figure 25.7e) can lead to several types of pulmonary aspergillosis. One type is allergic aspergillosis. Infected individuals may develop an immediate allergic response and suffer typical asthmatic attacks when exposed to fungal antigens on the conidiospores. In bronchopulmonary aspergillosis the major clinical manifestation of the allergic response is a bronchitis resulting from both type I and type III hypersensitivities (see figures 33.2 and 33.6). Although tissue invasion seldom occurs in bronchopulmonary aspergillosis, *Aspergillus* often can be cultured from the sputum. A most common manifestation of pulmonary involvement is the occurrence of colonizing aspergillosis, in which *Aspergillus* forms colonies within the lungs that develop into “fungus balls” called aspergillomas. These consist of a tangled mass of mycelia growing in a circumscribed area. From the pulmonary focus, the fungus may spread, producing disseminated aspergillosis in a variety of tissues and organs (figure 40.15). In patients whose resistance has been severely compromised, invasive aspergillosis may occur and fill the lung with fungal mycelia.

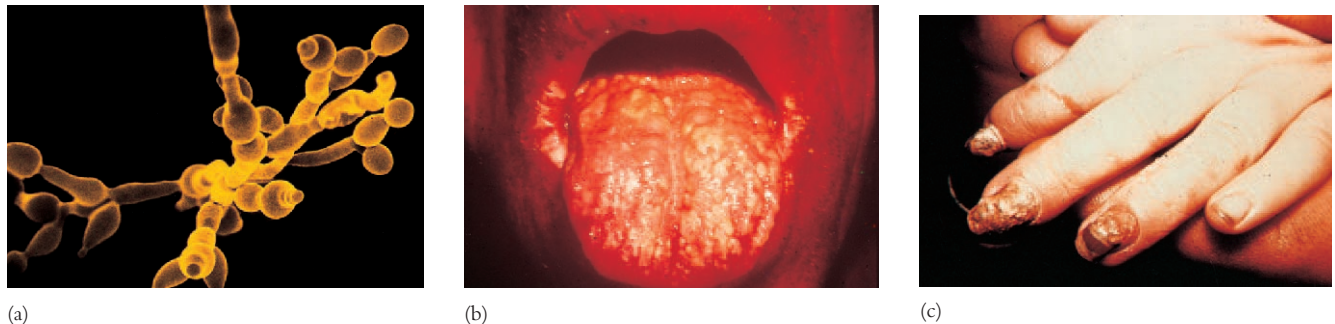


Figure 40.16 Opportunistic Mycoses Caused by *Candida albicans*. (a) Scanning electron micrograph of the yeast form ($\times 10,000$). Notice that some of the cells are reproducing by budding. (b) Thrush, or oral candidiasis, is characterized by the formation of white patches on the mucous membranes of the tongue and elsewhere in the oropharyngeal area. These patches form a pseudomembrane composed of spherical yeast cells, leukocytes, and cellular debris. (c) Paronychia and onychomycosis of the hands.

Box 40.1

The Emergence of Candidiasis

Written descriptions of oral lesions that were probably thrush date back to the early 1800s. In 1839 Bernard Langenbeck in Germany described the organism he found in oral lesions of a patient as “Typhus-Leichen” (typhus bodies). By 1841 Emil Berg established the fungal etiology of thrush by infecting healthy babies with what he called “aphthous membrane material.” In 1843 Charles Robin gave the organism its first name: *Oidium albicans*. Since then, more than 100 synonyms have been used for this fungus; of them all, *Candida albicans*, the name Roth Berkhout proposed in 1923, has persisted.

In 1861 Albert Zenker described the first well-documented case of systemic candidiasis. Historically, the most interesting period for candidiasis research coincided with the introduction of antibiotics. Since then there have been documented cases of this fungus involving all tissues and organs of the body, as well as an increase in the overall incidence of candidiasis. Some *Candida*-associated infections and diseases include arthritis, endophthalmitis, meningitis, myocarditis, myositis, and peritonitis. Besides the widespread use of antibiotics, other therapeutic and surgical procedures such as organ transplants and prostheses have been important in the expanding worldwide incidence of candidiasis.

Laboratory diagnosis of aspergillosis depends on identification, either by direct examination of pathological specimens or by isolation and characterization of the fungus. Successful therapy depends on treatment of the underlying disease so that host resistance increases. Treatment is with itraconazole.

Candidiasis is the mycosis caused by the dimorphic fungus *Candida albicans* (figure 40.16a). In contrast to the other pathogenic fungi, *C. albicans* is a member of the normal microbiota within the gastrointestinal tract, respiratory tract, vaginal area, and mouth (see figure 31.2). In healthy individuals *C. albicans* does not produce disease. Growth is suppressed by other microbiota. However, if anything upsets the normal microbiota, *Candida* may multiply rapidly and produce candidiasis. Recently *Candida* species have become important nosocomial pathogens. In some hospitals they may represent almost 10% of nosocomial bloodstream infections. Because *Candida* can be transmitted sex-

ually, it is also listed by the CDC as a sexually transmitted disease (see table 39.4).

No other mycotic pathogen produces as diverse a spectrum of disease in humans as does *C. albicans* (Box 40.1). Most infections involve the skin or mucous membranes. This occurs because *C. albicans* is a strict aerobe and finds such surfaces very suitable for growth. Cutaneous involvement usually occurs when the skin becomes overtly moist or damaged.

Oral candidiasis, or thrush (figure 40.16b), is a fairly common disease in newborns. It is seen as many small white flecks that cover the tongue and mouth. At birth, newborns do not have a normal microbiota in the oropharyngeal area. If the mother's vaginal area is heavily infected with *C. albicans*, the upper respiratory tract of the newborn becomes colonized during passage through the birth canal. Thrush occurs because growth of *C. albicans* cannot be inhibited by the other microbiota. Once the newborn has developed

its own normal oropharyngeal microbiota, thrush becomes uncommon. **Paronychia** and **onychomycosis** are associated with *Candida* infections of the subcutaneous tissues of the digits and nails, respectively (figure 40.16c). These infections usually result from continued immersion of the appendages in water.

Intertriginous candidiasis involves those areas of the body, usually opposed skin surfaces, that are warm and moist: axillae, groin, skin folds. **Napkin (diaper) candidiasis** is typically found in infants whose diapers are not changed frequently and therefore are not kept dry. **Candidal vaginitis** can result from diabetes, antibiotic therapy, oral contraceptives, pregnancy, or any other factor that compromises the female host. Normally the omnipresent lactobacilli (Döderlein's bacilli) can control *Candida* in this area by the low pH they create. However, if their numbers are decreased by any of the aforementioned factors, *Candida* may proliferate, causing a curdlike, yellow-white discharge from the vaginal area. *Candida* can be transmitted to males during intercourse and lead to **balanitis**; thus it also can be considered a sexually transmitted disease. Balanitis is a *Candida* infection of the male glans penis and occurs primarily in uncircumcised males. The disease begins as vesicles on the penis that develop into patches and are accompanied by severe itching and burning.

Diagnosis of candidiasis is difficult because (1) this fungus is a frequent secondary invader in diseased hosts, (2) a mixed microbiota is most often found in the diseased tissue, and (3) no completely specific immunologic procedures for the identification of *Candida* currently exist. There is no satisfactory treatment for candidiasis. Cutaneous lesions can be treated with topical agents such as sodium caprylate, sodium propionate, gentian violet, nystatin, miconazole, and trichomycin. Ketoconazole, amphotericin B, fluconazole, itraconazole, and flucytosine also can be used for systemic candidiasis.

Pneumocystis carinii are eucaryotic protists found in the lungs of a wide variety of mammals. The natural history of *P. carinii* is poorly understood, in large part because of a lack of a continuous in vitro culture system for its propagation. Although it was once considered a protozoan parasite, recent comparisons of rRNA and DNA sequences from several genes have shown that *P. carinii* is more closely related to fungi than protozoa.

The disease that this protist causes, **pneumocystis pneumonia** or ***Pneumocystis carinii* pneumonia (PCP)**, occurs almost exclusively in immunocompromised hosts. Extensive use of immunosuppressive drugs and irradiation for the treatment of cancers and following organ transplants accounts for the formidable prevalence rates noted recently. This pneumonia also occurs in more than 80% of AIDS patients. Both the organism and the disease remain localized in the lungs—even in fatal cases. Within the lungs *Pneumocystis* causes the alveoli to fill with a frothy exudate.

Laboratory diagnosis of pneumocystis pneumonia can be made definitively only by microscopically demonstrating the presence of the microorganisms in infected lung material or by a PCR analysis. Treatment is by means of oxygen therapy and ei-

ther a combination of trimethoprim (Bactrim) and sulfamethoxazole (Septra), atovaquone (Mepron), or trimetrexate (Neutrexin). Prevention and control is through prophylaxis with drugs in susceptible persons.

1. How are human fungal diseases categorized?
2. What are two piedrae that infect humans?
3. Briefly describe the major tineaes that occur in humans.
4. Describe the three types of subcutaneous mycoses that affect humans.
5. Why is *Histoplasma capsulatum* found in bird feces but not within the birds themselves?
6. Why are some mycotic diseases of humans called opportunistic mycoses?
7. What parts of the human body can be affected by *Candida* infections?
8. When is *Pneumocystis carinii* pneumonia likely to occur in humans?

40.2 Protozoan Diseases

Protozoa have become adapted to practically every type of habitat on the face of the earth, including the human body. Though fewer than 20 genera of protozoa cause disease in humans (**tables 40.3 and 40.4**), their impact is formidable. For example, there are over 150 million cases of malaria in the world each year. In tropical Africa alone, malaria is responsible each year for the deaths of more than a million children under the age of 14. It is estimated that there are at least 8 million cases of trypanosomiasis, 12 million cases of leishmaniasis, and over 500 million cases of amebiasis yearly. There is an increasing problem with *Cryptosporidium* and *Cyclospora* contamination of food and water supplies. More of our population is elderly, and a growing number of persons are immunosuppressed due to the human immunodeficiency virus, organ transplantation, or cancer chemotherapy. These populations are at increased risk for protozoan infections. The remainder of this chapter discusses some of these important protozoan diseases of humans.

Amebiasis

It is now accepted that two species of *Entamoeba* infect humans: the nonpathogenic *E. dispar* and the pathogenic *E. histolytica*. *E. histolytica* is responsible for **amebiasis (amebic dysentery)**. This very common parasite is endemic in warm climates where adequate sanitation and effective personal hygiene is lacking. Within the United States about 3,000 to 5,000 cases are reported annually. However, it is a major cause of parasitic death worldwide; about 500 million people are infected and as many as 100,000 die of amebiasis each year.

Infection occurs by ingestion of mature cysts. After excystation in the lower region of the small intestine, the metacyst di-

Table 40.3 Examples of Some Medically Important Protozoa

Phylum	Group	Pathogen	Disease
<i>Sarcomastigophora</i>	Amoebae	<i>Entamoeba histolytica</i> <i>Acanthamoeba</i> spp., <i>Naegleria fowleri</i>	Amebiasis, amebic dysentery Amebic meningoencephalitis
<i>Apicomplexa</i>	Coccidia	<i>Cryptosporidium parvum</i> <i>Cyclospora cayatanensis</i> <i>Isospora belli</i> <i>Toxoplasma gondii</i>	Cryptosporidiosis Cyclosporidiosis Isosporiasis Toxoplasmosis
<i>Ciliophora</i>	Ciliates	<i>Balantidium coli</i>	Balantidiasis
<i>Sarcomastigophora</i>	Blood and tissue flagellates	<i>Leishmania tropica</i> <i>L. braziliensis</i> <i>L. donovani</i> <i>Trypanosoma cruzi</i> <i>T. brucei gambiense</i> , <i>T. brucei rhodesiense</i>	Cutaneous leishmaniasis Mucocutaneous leishmaniasis Kala-azar (visceral leishmaniasis) American trypanosomiasis African sleeping sickness
<i>Sarcomastigophora</i>	Digestive and genital organ flagellates	<i>Giardia lamblia</i> <i>Trichomonas vaginalis</i>	Giardiasis Trichomoniasis
<i>Apicomplexa</i>	Sporozoa	<i>Plasmodium falciparum</i> , <i>P. malariae</i> , <i>P. ovale</i> , <i>P. vivax</i>	Malaria
<i>Microspora</i>	Microsporidia	<i>Encephalitozoon</i> , <i>Nosema</i> , <i>Vitta forma</i> , <i>Pleistophora</i> , <i>Enterocytozoon</i> , <i>Trachipleistophora</i> , <i>Microsporidium</i>	Microsporidiosis

Table 40.4 Examples of Some Recently
Recognized Human Protozoan Diseases

Year	Protozoan	Disease
1976	<i>Cryptosporidium parvum</i>	Acute and chronic diarrhea, cryptosporidiosis
1985	<i>Enterocytozoon bienersi</i>	Diarrhea, microsporidiosis
1986	<i>Cyclospora cayatanensis</i>	Persistent diarrhea
1991	<i>Encephalitozoon hellem</i>	Conjunctivitis, disseminated microsporidiosis
1991	<i>Babesia</i> spp.	Atypical babesiosis
1993	<i>Encephalitozoon cuniculi</i>	Disseminated microsporidiosis
1996	<i>Trachipleistophora hominis</i>	Disseminated microsporidiosis
1998	<i>Brachiola vesicularum</i>	Myositis

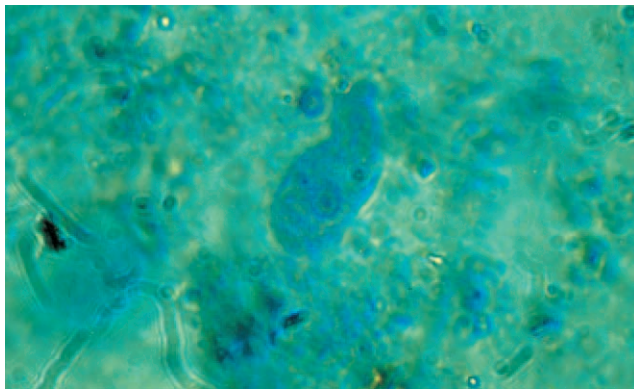
vides rapidly to produce eight small trophozoites (figure 40.17). These trophozoites move to the large intestine where they can invade the host tissue, live as commensals in the lumen of the intestine, or undergo encystation.

If the infective trophozoites invade the intestinal tissues, they multiply rapidly and spread laterally, while feeding on erythrocytes, bacteria, and yeasts. The invading trophozoites destroy the epithelial lining of the large intestine by producing cysteine proteinases. Cysteine proteinases possibly are a virulence factor of *E. histolytica* and may play a role in intestinal invasion by degrading

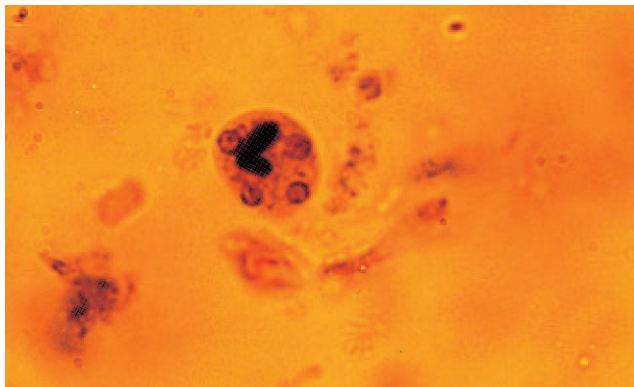
extracellular matrix and circumventing the host immune response through cleavage of secretory immunoglobulin A (sIgA), IgG, and complement factors. Cysteine proteinases are encoded by at least seven genes, several of which are found in *E. histolytica* but not *E. dispar*. Lesions (ulcers) are characterized by minute points of entry into the mucosa and extensive enlargement of the lesion after penetration into the submucosa. *E. histolytica* also may invade and produce lesions in extraintestinal foci, especially the liver, to cause hepatic amebiasis. However, all extraintestinal amebic lesions are secondary to the ones established in the large intestine.

The symptoms of amebiasis are highly variable, ranging from an asymptomatic infection to fulminating dysentery, exhaustive diarrhea accompanied by blood and mucus, appendicitis, and abscesses in the liver, lungs, or brain.

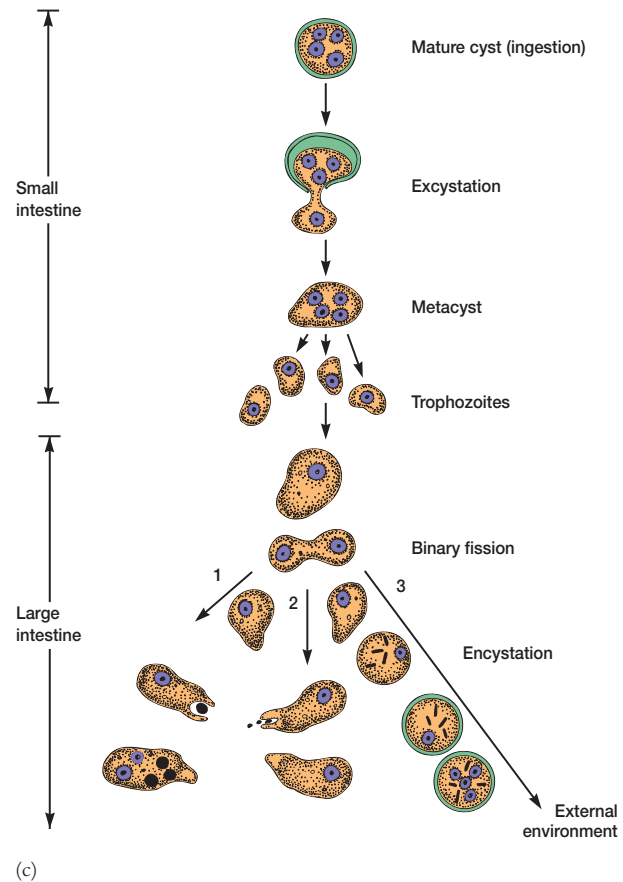
Laboratory diagnosis of amebiasis is based on finding trophozoites in fresh warm stools and cysts in ordinary stools. Serological testing also should be done. The therapy for amebiasis is complex and depends on the location of the infection within the host and the host's condition. Asymptomatic carriers that are passing cysts should always be treated with iodoquinol or paromomycin because they represent the most important reservoir of the parasite in the population. In symptomatic intestinal amebiasis, metronidazole (Flagyl) or iodoquinol (Yodoxin) are the drugs of choice. Prevention and control of amebiasis is achieved by avoiding water or food that might be contaminated with human feces in endemic areas. Viable cysts in water can be destroyed by hyperchlorination or iodination.



(a)



(b)



(c)

Figure 40.17 Amebiasis Caused by *Entamoeba histolytica*. (a) Light micrographs of a trophozoite ($\times 1,000$) and (b) a cyst ($\times 1,000$). (c) Life cycle. Infection occurs by the ingestion of a mature cyst of the parasite. Excystation occurs in the lower region of the small intestine and the metacyst rapidly divides to give rise to eight small trophozoites (only four are shown). These enter the large intestine, undergo binary fission, and may (1) invade the host tissues, (2) live in the lumen of the large intestine without invasion, or (3) undergo encystation and pass out of the host in the feces.

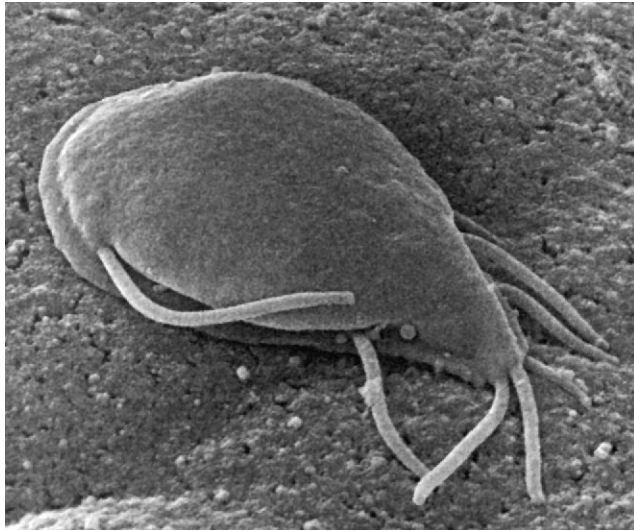
Cryptosporidiosis

The first case of human **cryptosporidiosis** was reported in 1976 (table 40.4). The protozoan responsible was identified as *Cryptosporidium parvum*. In 1993 *C. parvum* contaminated the Milwaukee, Wisconsin, water supply and caused severe diarrheal disease in about 400,000 individuals, the largest recognized outbreak of waterborne illness in U.S. history. *Cryptosporidium* ("hidden spore cysts") is found in about 90% of sewage samples, in 75% of river waters, and in 28% of drinking waters.

C. parvum is a common coccidian apicomplexan parasite found in the intestine of many birds and mammals. When these animals defecate, oocysts are shed into the environment. If a human ingests food or water that is contaminated with the oocysts, excystation occurs within the small intestine and sporozoites enter epithelial cells and develop into merozoites. Some of the merozoites subsequently undergo sexual reproduction to produce zygotes, and

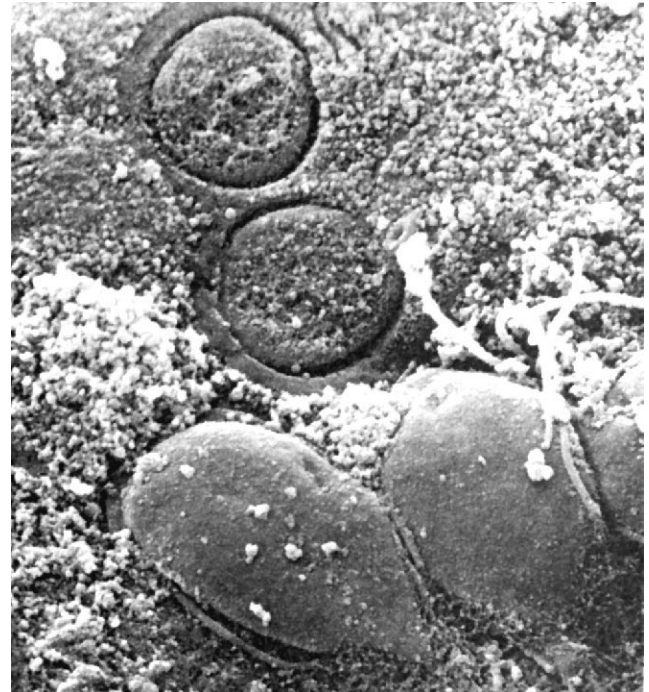
the zygotes differentiate into thick-walled oocysts. Oocyst release into the environment begins the life cycle again. A major problem for public health arises from the fact that the oocysts are only 4 to 6 μm in diameter, much too small to be easily removed by the sand filters used to produce drinking water. *Cryptosporidium* also is extremely resistant to disinfectants such as chlorine. The problem is made worse by the low infectious dose, around 10 to 100 oocysts, and the fact that the oocysts may remain viable for 2 to 6 months in a moist environment. [Water purification \(pp. 651–53\).](#)

The incubation period for cryptosporidiosis ranges from 5 to 28 days. Diarrhea, which characteristically may be choleralike, is the most common symptom. Other symptoms include abdominal pain, nausea, fever, and fatigue. The pathogen is routinely diagnosed by fecal concentration and acid-fast stained smears; rapid EIA and DFA techniques also have been developed. No chemotherapy is available and patients are simply rehydrated. Although the



(a)

Figure 40.18 Giardiasis. (a) *Giardia lamblia* adhering to the epithelium by its sucking disk; scanning electron micrograph. (b) Upon detachment from the epithelium, the protozoa often leave clear impressions on the microvillus surface (upper circles); scanning electron micrograph.



(b)

disease usually is self-limiting in healthy individuals, patients that have late-stage AIDS or are immunocompromised in other ways may develop prolonged, severe, and life-threatening diarrhea.

Freshwater Amoeba Diseases

Free-living amoebae of the genera *Naegleria* and *Acanthamoeba* are facultative parasites responsible for causing **primary amebic meningoencephalitis** (see table 39.2) in humans. They are among the most common protozoa found in freshwater and moist soil. In addition, several *Acanthamoeba* spp. are known to infect the eye, causing a chronically progressive ulcerative *Acanthamoeba* **keratitis**, inflammation of the cornea, which may result in blindness. Wearers of soft contact lenses may be predisposed to this infection and should take care to prevent contamination of their lens cleaning and soaking solutions. Diagnosis of these infections is by demonstration of the amoebae in clinical specimens. Most freshwater amoebae are resistant to commonly used antibiotic agents. These amoebae are reported in fewer than 100 human disease cases annually in the United States although the incidence (especially of *Acanthamoeba* keratitis) is likely higher.

Giardiasis

Giardia lamblia [syn., *G. duodenalis*, *G. intestinalis*] is a flagellated protozoan (figure 40.18a) that causes the very common intestinal disease **giardiasis**. (It was discovered by van Leeuwen-

hoek in 1681 when he examined his own stools.) *G. lamblia* is worldwide in distribution, and it affects children more seriously than it does adults. In the United States this protozoan is the most common cause of epidemic waterborne diarrheal disease (about 30,000 cases yearly). Approximately 7% of the population are healthy carriers and shed cysts in their feces. *G. lamblia* is endemic in child day-care centers in the United States, with estimates of 5 to 15% of diapered children being infected.

Transmission is most frequent by cyst-contaminated water supplies. Epidemic outbreaks have been recorded in wilderness areas, suggesting that humans may be infected from “clean water” with *Giardia* harbored by rodents, deer, cattle, or household pets. This implies that human infections also can be a zoonosis (see table 37.1). As many as 200 million humans may be infected worldwide.

Following ingestion, the cysts undergo excystation in the duodenum, forming trophozoites. The trophozoites inhabit the upper portions of the small intestine, where they attach to the intestinal mucosa by means of their sucking disks (figure 40.18a). The ability of the trophozoites to adhere to the intestinal epithelium accounts for the fact that they are rarely found in stools. It is thought that the trophozoites feed on mucous secretions and reproduce to form such a large population that they interfere with nutrient absorption by the intestinal epithelium.

Giardiasis varies in severity, and asymptomatic carriers are common. The disease can be acute or chronic. Acute giardiasis is characterized by severe diarrhea, epigastric pain, cramps, voluminous flatulence (“passing gas”), and anorexia. Chronic giardiasis

Box 40.2

A Brief History of Malaria

No other single infectious disease has had the impact on humans that malaria has. The first references to its periodic fever and chills can be found in early Chaldean, Chinese, and Hindu writings. In the late fifth century B.C., Hippocrates described certain aspects of malaria. In the fourth century B.C., the Greeks noted an association between individuals exposed to swamp environments and the subsequent development of periodic fever and enlargement of the spleen (splenomegaly). In the seventeenth century the Italians named the disease *mal'aria* (bad air) because of its association with the ill-smelling vapors from the swamps near Rome. At about the same time, the bark of the quina-quina (cinchona) tree of South America was used to treat the intermittent fevers, although it was not until the mid-nineteenth century that quinine was identified as the active alkaloid. The major epidemiological breakthrough came in 1880, when French army surgeon Charles Louis Alphonse Laveran observed exflagellated gametocytes in fresh blood. Five years later the Italian histologist Camillo Golgi observed the multiplication of the asexual blood forms. In the late 1890s Patrick Manson postulated that malaria was transmitted by mosquitoes. Sir Ronald Ross, a British army surgeon in the Indian Medical Service, subsequently observed developing plasmodia in the intestine of mosquitoes, supporting Manson's theory. Using birds as experimental models, Ross definitively established the major features of the life cycle of *Plasmodium* and received the Nobel Prize in 1902.

Human malaria is known to have contributed to the fall of the ancient Greek and Roman empires. Troops in both the United States Civil War and the Spanish-American War were severely incapacitated by the

disease. More than 25% of all hospital admissions during these wars were malaria patients. During World War II malaria epidemics severely threatened both the Japanese and Allied forces in the Pacific. The same can be said for the military conflicts in Korea and Vietnam.

In the twentieth century efforts have been directed toward understanding the biochemistry and physiology of malaria, controlling the mosquito vector, and developing antimalarial drugs. In the 1960s it was demonstrated that resistance to *P. falciparum* among West Africans was associated with the presence of hemoglobin-S (Hb-S) in their erythrocytes. Hb-S differs from normal hemoglobin-A by a single amino acid, valine, in each half of the Hb molecule. Consequently these erythrocytes—responsible for sickle cell disease—have a low binding capacity for oxygen. Because the malarial parasite has a very active aerobic metabolism, it cannot grow and reproduce within these erythrocytes.

In 1955 the World Health Organization began a worldwide malarial eradication program that finally collapsed by 1976. Among the major reasons for failure were the development of resistance to DDT by the mosquito vectors and the development of resistance to chloroquine by strains of *Plasmodium*. Currently scientists are exploring new approaches, such as the development of vaccines and more potent drugs. For example, in 1984 the gene encoding the sporozoite antigen was cloned, permitting the antigen to be mass-produced by genetic engineering techniques. Overall, no greater achievement for molecular biology could be imagined than the control of malaria—a disease that has caused untold misery throughout the world since antiquity and remains one of the world's most serious infectious diseases.

is characterized by intermittent diarrhea, with periodic appearance and remission of symptoms.

Laboratory diagnosis is based on the identification of trophozoites—only in the severest of diarrhea—or cysts in stools. A commercial ELISA test is also available for the detection of *G. lamblia* antigen in stool specimens. Quinacrine hydrochloride (Atabrine) and metronidazole (Flagyl) are the drugs of choice for adults, and furazolidone is used for children because it is available in a pleasant-tasting liquid suspension. Prevention and control involves proper treatment of community water supplies, especially the use of slow sand filtration (*see section 29.5*) because the cysts are highly resistant to chlorine treatment.

Malaria

The most important human parasite among the sporozoa is *Plasmodium*, the causative agent of **malaria (Box 40.2)**. It has been estimated that more than 150 million people are infected, and about one million die annually of malaria in Africa alone. About 1,000 cases are reported each year in the United States, divided between returning U.S. travelers and non-U.S. citizens.

Human malaria is caused by four species of *Plasmodium*: *P. falciparum*, *P. malariae*, *P. vivax*, and *P. ovale*. The life cycle of

P. vivax is shown in **figure 40.19**. The parasite first enters the bloodstream through the bite of an infected female *Anopheles* mosquito. As she feeds, the mosquito injects a small amount of saliva containing an anticoagulant along with small haploid sporozoites (*see figure 27.3c*). The sporozoites in the bloodstream immediately enter hepatic cells of the liver. In the liver they undergo multiple asexual fission (schizogony) and produce merozoites. After being released from the liver cells, the merozoites attach to erythrocytes and penetrate these cells.

Once inside the erythrocyte, the *Plasmodium* begins to enlarge as a uninucleate cell termed a trophozoite. The trophozoite's nucleus then divides asexually to produce a schizont that has 6 to 24 nuclei. The schizont divides and produces mononucleated merozoites. Eventually the erythrocyte lyses, releasing the merozoites into the bloodstream to infect other erythrocytes. This erythrocytic stage is cyclic and repeats itself approximately every 48 to 72 hours or longer, depending on the species of *Plasmodium* involved. This sudden release of merozoites, toxins, and erythrocyte debris triggers an attack of the chills and fever, so characteristic of malaria. Occasionally, merozoites differentiate into macrogametocytes and microgametocytes, which do not rupture the erythrocyte. When these are ingested by a mosquito, they develop into female and male gametes, respectively. In the mosquito's gut the

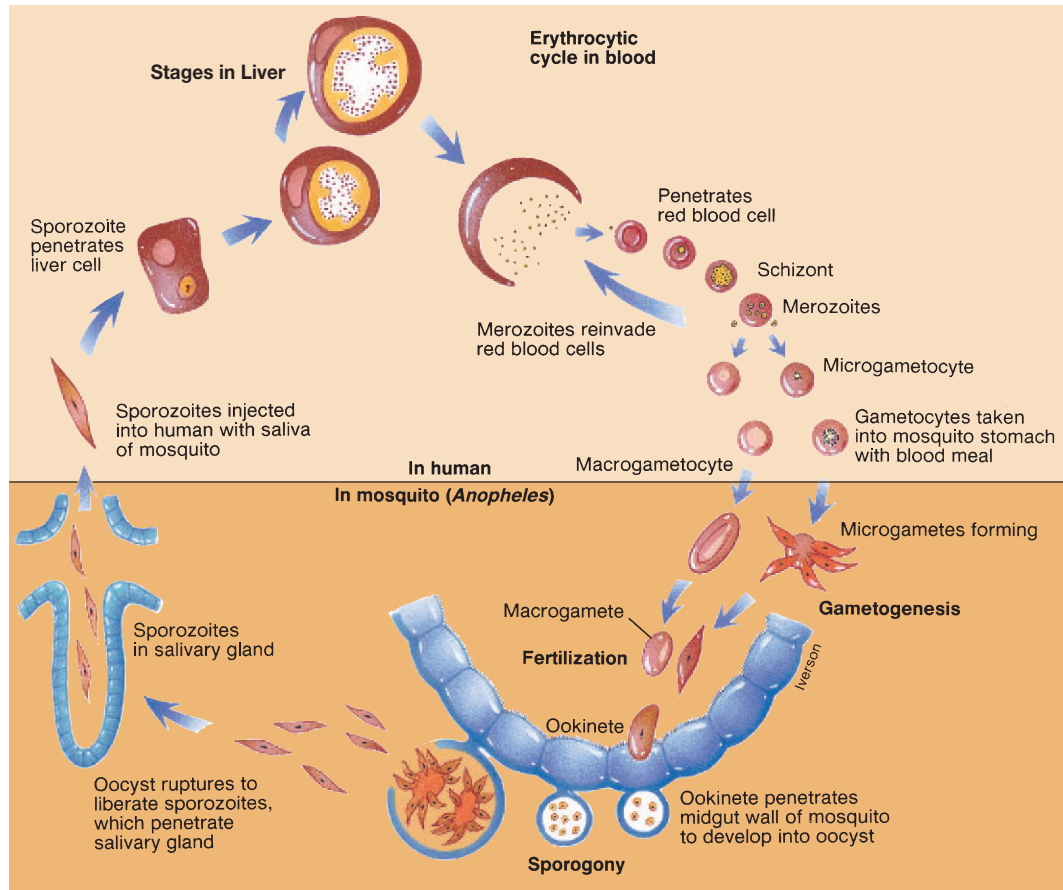


Figure 40.19 Malaria. Life cycle of *Plasmodium vivax*. See text for details.

infected erythrocytes lyse and the gametes fuse to form a diploid zygote called the ookinete. The ookinete migrates to the mosquito's gut wall, penetrates, and forms an oocyst on its outer surface. In a process called sporogony, the oocyst undergoes meiosis and forms sporozoites that migrate to the salivary glands of the mosquito. The cycle is now complete, but when the mosquito bites another human host, the cycle begins anew.

The pathological changes caused by malaria involve not only the erythrocytes but also the spleen and other visceral organs. Classic symptoms first develop with the synchronized release of merozoites and erythrocyte debris into the bloodstream, resulting in the malarial paroxysms—shaking chills, then burning fever followed by sweating. It may be that the fever and chills are caused partly by a malarial toxin that induces macrophages to release TNF- α and interleukin-1. Several of these paroxysms constitute an attack. After one attack there is a remission that lasts from a few weeks to several months. Then there is a relapse. Between paroxysms, the patient feels normal. Anemia can result from the loss of

erythrocytes, and the spleen and liver often hypertrophy. Children and nonimmune individuals can die of cerebral malaria.

Diagnosis of malaria is made by demonstrating the presence of parasites within Wright- or Giemsa-stained erythrocytes (**figure 40.20**). When blood smears are negative, serological testing can establish a diagnosis of malaria in individuals. Specific recommendations for treatment are region dependent. Treatment includes administration of chloroquine, amodiaquine, or mefloquine. These suppressant drugs are effective in eradicating erythrocytic asexual stages. Primaquine has proved satisfactory in eradicating the exoerythrocytic stages. However, because resistance to these drugs is occurring rapidly, more expensive drug combinations are being used. One example is Fansidar, a combination of pyrimethamine and sulfadoxine. It is worth noting that individuals who are traveling to areas where malaria is endemic (**figure 40.21**) should receive chemoprophylactic treatment with chloroquine. Efforts to develop a vaccine are under way.

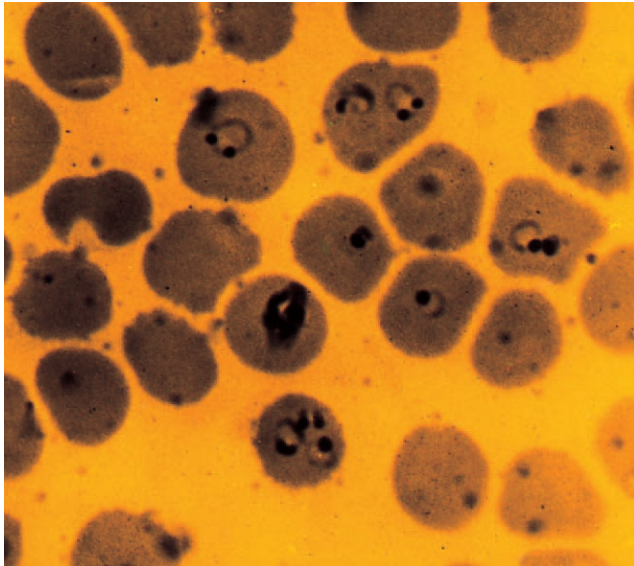


Figure 40.20 Malaria: Erythrocytic Cycle. Trophozoites of *P. falciparum* in circulating erythrocytes; light micrograph ($\times 1,100$). The young trophozoites resemble small rings resting in the erythrocyte cytoplasm.

Hemoflagellate Diseases

The flagellated protozoa that are transmitted by the bites of infected arthropods and infect the blood and tissues of humans are called **hemoflagellates**. There are two major groups of pathogens: the leishmanias and the trypanosomes.

Leishmaniasis

Leishmanias are flagellated protozoa that cause a group of human diseases collectively called **leishmaniasis**. The primary reservoirs of these parasites are canines and rodents. All species of *Leishmania* use sand flies such as those of the genera *Lutzomyia* and *Phlebotomus* as intermediate hosts. The leishmanias are transmitted from animals to humans or between humans by these sand flies. When an infected sand fly takes a human blood meal, it introduces flagellated promastigotes into the skin of the definitive host. Within the skin the promastigotes are engulfed by macrophages, multiply by binary fission and form small nonmotile cells called amastigotes. These destroy the host cell, and are engulfed by other macrophages in which they continue to develop and multiply.

Leishmania braziliensis, which has an extensive distribution in forest regions of tropical America, causes mucocutaneous leishmaniasis (espundia) (**figure 40.22a**). The disease produces

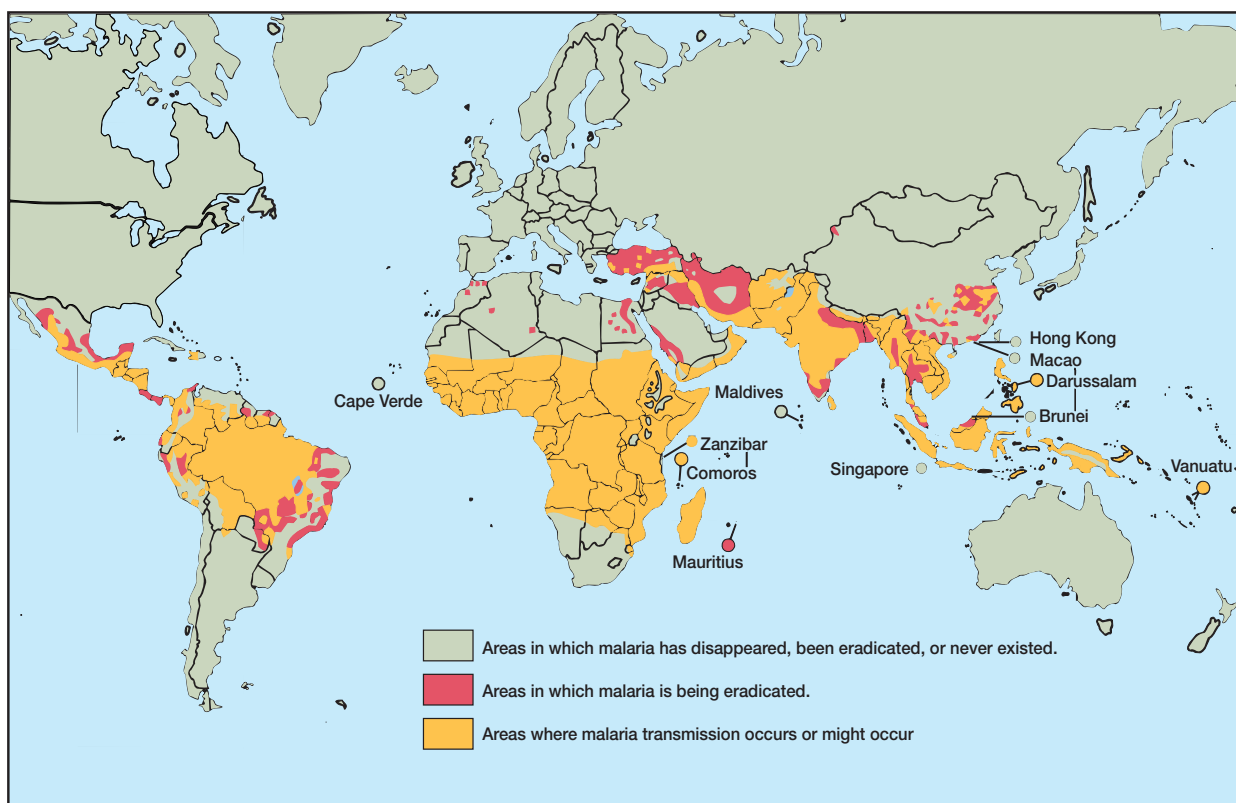
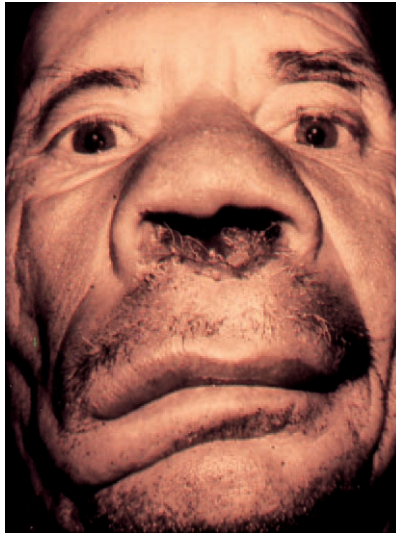


Figure 40.21 Geographic Distribution of Malaria. Notice that malaria is endemic around the equator. Source: Data from the World Health Statistics Quarterly, 41:69, 1988, World Health Organization, Switzerland.



(a)



(b)

Figure 40.22 Leishmaniasis. (a) A person with mucocutaneous leishmaniasis, which has destroyed the nasal septum and deformed the nose and lips. (b) A person with diffuse cutaneous leishmaniasis.

lesions involving the mouth, nose, throat, and skin and results in extensive scarring and disfigurement.

Leishmania donovani is endemic in large areas within northern China, eastern India, the Mediterranean countries, the Sudan, and Latin America. It produces visceral leishmaniasis (kala-azar). The disease involves the reticuloendothelial system and often results in intermittent fever and enlargement of the spleen and liver. Individuals who recover develop a permanent immunity.

Leishmania tropica and *L. mexicana* occur in the more arid regions of the Eastern Hemisphere and cause cutaneous leishmaniasis. *L. mexicana* is also found in the Yucatan Peninsula (Mexico) and has been reported as far north as Texas. In this disease a relatively small red papule forms at the site of each insect bite—the inoculation site. These papules are frequently found on the face and ears. They eventually develop into crustated ulcers (figure 40.22b). Healing occurs with scarring and a permanent immunity.

Laboratory diagnosis of leishmaniasis is based on finding the parasite within infected macrophages in stained smears from lesions or infected organs. Treatment includes the pentavalent antimonial compounds (Pentostam, Glucantime). Vector and reservoir control and aggressive epidemiological surveillance are the best options for prevention and containment of this disease.

Trypanosomiasis

Another group of flagellated protozoa called **trypanosomes** (see figure 27.3a) cause the aggregate of diseases termed **trypanosomiasis**. *Trypanosoma brucei gambiense*, found in the rain forests of west and central Africa, and *T. brucei rhodesiense*, found in the upland savannas of east Africa, cause African trypanosomiasis. Reservoirs for these trypanosomes are domestic cattle and wild animals, within which the parasites cause severe malnutrition. Both species use tsetse flies as intermediate hosts. The parasites are transmitted through the bite of the fly to humans. Once the protozoa enter the bloodstream, they begin to multiply. They

cause interstitial inflammation and necrosis within the lymph nodes and small blood vessels of the brain and heart. In *T. brucei rhodesiense* infection the disease develops so rapidly that infected individuals often die within a year. In *T. brucei gambiense* infection the parasites invade the central nervous system, where necrotic damage causes a variety of nervous disorders, including the characteristic sleeping sickness. The name sleeping sickness is derived from the lethargy of the host—characteristically lying prostrate, drooling from the mouth, and insensitive to pain. Usually the victim dies in 2 to 3 years. Trypanosomiasis is such a problem in parts of Africa that millions of square miles are not fit for human habitation.

T. cruzi causes **American trypanosomiasis (Chagas' disease)**, which occurs in the tropics and subtropics of continental America. The parasite uses the triatomid (kissing) bug as a vector. As the triatomid bug takes a blood meal, the parasites are discharged in the insect's feces. Some trypanosomes enter the bloodstream through the wound and invade the liver, spleen, lymph nodes, and central nervous system, where they multiply and destroy the parasitized cells. In some parts of Latin America, a high percentage of heart disease is due to parasitized cardiac cells.

Trypanosomiasis is diagnosed by finding motile parasites in fresh blood and by serological testing. Treatment for African trypanosomiasis uses suramin and pentamidine for non-nervous-system involvement and melarsoprol when the nervous system is involved. Currently there is no drug suitable for Chagas' disease, although nifurtimox (Lampit) and benznidazole have shown some value. Vaccines are not useful because the parasite is able to change its protein coat and evade the immunologic response.

Toxoplasmosis

Toxoplasmosis is a disease caused by the protozoan *Toxoplasma gondii*. This apicomplexan protozoan has been found in nearly all animals and most birds; cats are the definitive host and required

for completion of the sexual cycle. Animals shed oocysts in the feces; the oocysts enter another host by way of the nose or mouth; and the parasites colonize the intestine. Toxoplasmosis also can be transmitted by the ingestion of raw or undercooked meat, congenital transfer, blood transfusion, or a tissue transplant. Originally toxoplasmosis gained public notice when it was discovered that in pregnant women the protozoan might also infect the fetus, causing serious congenital defects or death.

Most cases of toxoplasmosis are asymptomatic. Adults usually complain of an “infectious mononucleosis-like” syndrome. In immunoincompetent or immunosuppressed individuals, it frequently results in fatal disseminated disease with a heavy cerebral involvement.

Acute toxoplasmosis is usually accompanied by lymph node swelling (lymphadenopathy) with reticular cell enlargement or hyperplasia. Pulmonary necrosis, myocarditis, and hepatitis caused by tissue necrosis are common. Retinitis (inflammation of the retina of the eye) is associated with the necrosis due to the proliferation of the parasite within retinal cells. Currently toxoplasmosis has become a major cause of death in AIDS patients from a unique encephalitis with necrotizing lesions accompanied by inflammatory infiltrates. It continues to cause more than 3,000 congenital infections per year in the United States.

Laboratory diagnosis of toxoplasmosis is by serological tests. Epidemiologically, toxoplasmosis is ubiquitous in all higher animals. Treatment of toxoplasmosis is with a combination of pyrimethamine (Daraprim) and sulfadiazine. Prevention and control requires minimizing exposure by the following: avoiding eating raw meat and eggs, washing hands after working in the soil, cleaning cat litterboxes daily, keeping personal cats indoors if possible, and feeding them commercial food.

Trichomoniasis

Trichomoniasis is a sexually transmitted disease (see table 39.4) caused by the protozoan flagellate *Trichomonas vaginalis* (figure 40.23). It is one of the most common sexually transmitted diseases, with an estimated 7 million cases annually in the United States and 180 million cases annually worldwide. In response to the parasite, the body accumulates leukocytes at the site of the infection. In females this usually results in a profuse purulent vaginal discharge that is yellowish to light cream in color and characterized by a disagreeable odor. The discharge is accompanied by itching. Males are generally asymptomatic because of the trichomonocidal action of prostatic secretions; however, at times a burning sensation occurs during urination. Diagnosis is made in females by microscopic examination of the discharge and identification of the parasite. Infected males will demonstrate the para-



Figure 40.23 Trichomoniasis. *Trichomonas vaginalis*, showing the characteristic undulating membranes and flagella; scanning electron micrograph ($\times 12,000$).

site in semen or urine. Treatment is by administration of metronidazole (Flagyl).

1. How do infections caused by *Entamoeba histolytica* occur?
2. What is the most common cause of epidemic waterborne diarrheal disease?
3. Describe in detail the life cycle of the malarial parasite.
4. What protozoa are represented within the hemoflagellates? What diseases do they cause?
5. In what two ways does *Toxoplasma* affect human health?
6. How would you diagnose trichomoniasis in a female? In a male?

Summary

- Human fungal diseases, or mycoses, can be divided into five groups according to the level and mode of entry into the host. These are the superficial, cutaneous, subcutaneous, systemic, and opportunistic mycoses (**table 40.1**).
- Superficial mycoses of the hair shaft are collectively called *piedras*. Two major types are black *piedra* (**figure 40.1**) and white *piedra*. *Tinea versicolor* is a third common superficial mycosis.
- The cutaneous fungi that parasitize the hair, nails, and outer layer of the skin are called dermatophytes, and their infections are termed dermatophytoses, ringworms, or *tineas*. At least seven types can occur in humans: *tinea barbae* (ringworm of the beard; **figure 40.2**), *tinea capitis* (ringworm of the scalp; **figure 40.3**), *tinea corporis* (ringworm of the body; **figure 40.4**), *tinea cruris* (ringworm of the groin; **figure 40.5**), *tinea pedis* (ringworm of the feet; **figure 40.6**), *tinea manuum* (ringworm of the hands), and *tinea unguium* (ringworm of the nails; **figure 40.7**).
- The dermatophytes that cause the subcutaneous mycoses are normal saprophytic inhabitants of soil and decaying vegetation. Three types of subcutaneous mycoses can occur in humans: chromoblastomycosis (**figure 40.8**), maduromycosis (**figure 40.9**), and sporotrichosis (**figure 40.10**).
- Most systemic mycoses that occur in humans are acquired by inhaling the spores from the soil where the free-living fungi are found. Four types can occur in humans: blastomycosis (**figure 40.11**), coccidioidomycosis, cryptococcosis, and histoplasmosis.
- An opportunistic organism is one that is generally harmless in its normal environment but that can become pathogenic in a compromised host. The most important opportunistic mycoses affecting humans include systemic aspergillosis (**figure 40.15**), candidiasis (**figure 40.16b**), and *Pneumocystis carinii* pneumonia.
- Protozoa are responsible for some of the most serious human diseases that affect hundreds of millions of people worldwide (**table 40.3**).
- Entamoeba histolytica* is the amoeboid protozoan responsible for amebiasis. This is a very common disease in warm climates throughout the world. It is acquired when one ingests the cysts with contaminated food or water (**figure 40.17**).
- Cryptosporidium parvum* is a common coccidial apicomplexan parasite that causes severe diarrheal disease. It is acquired from contaminated food or water.
- Giardia lamblia* is a flagellated protozoan that causes the common intestinal disease giardiasis (**figure 40.18**). This disease is distributed throughout the world, and in the United States it is the most common cause of waterborne diarrheal disease.
- The most important human parasite among the sporozoa is *Plasmodium*, the causative agent of malaria (**figure 40.19**). Human malaria is caused by four species of *Plasmodium*: *P. falciparum*, *P. vivax*, *P. malariae*, and *P. ovale*.
- The flagellated protozoa that are transmitted by arthropods and infect the blood and tissues of humans are called hemoflagellates. Two major groups occur: the leishmanias, which cause the diseases collectively termed leishmaniasis (**figure 40.22**), and the trypanosomes, which cause trypanosomiasis.
- Toxoplasmosis is a disease caused by the protozoan *Toxoplasma gondii*. It is one of the major causes of death in AIDS patients.
- Trichomoniasis is a sexually transmitted disease caused by the protozoan flagellate *Trichomonas vaginalis* (**figure 40.23**).

Key Terms

- | | | |
|--|---|--|
| amebiasis (amebic dysentery) 950 | hemoflagellate 956 | primary amebic meningoencephalitis 953 |
| American trypanosomiasis (Chagas' disease) 957 | histoplasmosis 947 | ringworm 943 |
| aspergillosis 948 | intertriginous candidiasis 950 | sporotrichosis 945 |
| athlete's foot 944 | jock itch 944 | thrush 949 |
| balanitis 950 | keratitis 953 | tinea 943 |
| black piedra 943 | leishmania 956 | tinea barbae 943 |
| blastomycosis 946 | leishmaniasis 956 | tinea capitis 943 |
| candidal vaginitis 950 | maduromycosis 945 | tinea corporis 943 |
| candidiasis 949 | malaria 954 | tinea cruris 944 |
| chromoblastomycosis 945 | medical mycology 942 | tinea manuum 944 |
| coccidioidomycosis 946 | mycosis 942 | tinea pedis 944 |
| compromised host 948 | napkin (diaper) candidiasis 950 | tinea unguium 944 |
| cryptococcosis 947 | onychomycosis 950 | tinea versicolor 943 |
| cryptosporidiosis 952 | opportunistic microorganism 948 | toxoplasmosis 957 |
| dermatophyte 943 | oral candidiasis 949 | trichomoniasis 958 |
| dermatomycosis 943 | paronychia 950 | trypanosome 957 |
| eumycotic mycetoma 945 | piedra 943 | trypanosomiasis 957 |
| extracutaneous sporotrichosis 945 | <i>Pneumocystis carinii</i> pneumonia (PCP) 950 | white piedra 943 |
| giardiasis 953 | pneumocystis pneumonia 950 | |

Questions for Thought and Review

- Briefly describe each of the major or most common fungal and protozoan diseases in terms of its causative agent, signs and symptoms, the course of infection, mechanism of pathogenesis, epidemiology, and prevention and/or treatment.
- There are no antibiotics—nothing equivalent to penicillin or streptomycin—to control fungi in the human body. Recalling the mechanisms by which antibiotics affect bacteria, why do you think there has been no similar success in the control of fungal pathogens?
- Give several reasonable explanations why most fungal diseases in humans are not contagious.
- What factors determine one's susceptibility to fungal diseases?
- What is the relationship between AIDS and mycotic and protozoan diseases?
- Why do most protozoan diseases occur in the tropics?
- Give several reasons why malaria is still one of the most serious of all infectious diseases that affect humans.
- What morphological property is the most important in the identification of fungi?
- Compared to fungal parasites, how are protozoan parasites transmitted?
- What are dermatomycoses?

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Critical Thinking Questions

- Compare and contrast treatment of diseases caused by fungi with those caused by viruses or bacteria.
 - What is one distinct feature of fungi that could be exploited for antibiotic therapy?
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PART XI

Food and Industrial Microbiology

Chapter 41
Microbiology of Food

Chapter 42
Industrial Microbiology
and Biotechnology

CHAPTER 41

Microbiology of Food



Large tanks used for wine production. Fermentations can be carried out in such open-air units in temperate regions. After completion of the fermentation, the fresh wine will be transferred to barrels for storage and aging.

Outline

- 41.1 Microorganism Growth in Foods 964
 - Intrinsic Factors 964
 - Extrinsic Factors 965
- 41.2 Microbial Growth and Food Spoilage 966
- 41.3 Controlling Food Spoilage 970
 - Removal of Microorganisms 970
 - Low Temperature 970
 - High Temperature 970
 - Water Availability 971
 - Chemical-Based Preservation 971
 - Radiation 972
 - Microbial Product-Based Inhibition 972
- 41.4 Food-Borne Diseases 973
 - Food-Borne Infection 973
 - Food-Borne Intoxications 975
- 41.5 Detection of Food-Borne Pathogens 976
- 41.6 Microbiology of Fermented Foods 978
 - Fermented Milks 978
 - Cheese Production 979
 - Meat and Fish 982
 - Production of Alcoholic Beverages 982
 - Production of Breads 984
 - Other Fermented Foods 985
- 41.7 Microorganisms as Foods and Food Amendments 986

Concepts

1. Foods often provide an ideal environment for microbial survival and growth.
2. Microbial growth in foods involves successional changes, with intrinsic, or food-related, and extrinsic, or environmental, factors interacting with the microbial community over time.
3. Food spoilage is a major problem in all societies. This can occur at any point in the course of food production, transport, storage, or preparation. Food-borne toxins are of increasing concern, especially with increases in international shipments and extended storage of food products before use. Growth of fungi can result in the synthesis of toxins such as aflatoxins, fumonisins, and ergot alkaloids. Algal-derived toxins can be transmitted to humans through freshwater and marine-derived food products.
4. Foods can be preserved by physical, chemical, and biological processes. Refrigeration does not significantly reduce microbial populations but only retards spoilage. Pasteurization results in a pathogen-free product with a longer shelf life.
5. Chemicals can be added to foods to control microbial growth. Such chemicals include sugar, salt (decreasing water availability), and many organic chemicals that affect specific groups of microorganisms. Microbial products, such as bacteriocins, can be added to foods to control spoilage organisms.
6. Foods can transmit a wide range of diseases to humans. In food infection the food serves as a vehicle for the transfer of the pathogen to the consumer, in whom the pathogen grows and causes disease. With food intoxication the microorganisms grow in the food and produce toxins that can then affect the consumer.

Concepts *(Continued)*

7. New pathogens continue to be discovered. This may be due to an increased ability to detect these pathogens as well as to better reporting and monitoring of disease occurrence.
8. Foods that are consumed raw, such as sprouts, shellfish/finfish, and fruits, are increasingly popular. There is a risk of disease transmission if care is not taken in production, storage, and transport of these perishable foods. Without adequate care, major disease outbreaks can occur because foods travel around the globe in extremely short times.
9. Detection of food-borne pathogens is carried out using classic culture techniques, as well as immunological and molecular procedures. Modern techniques make it possible to quickly link food-borne disease occurrences to common pathogen sources. With kit-type rapid diagnostic systems, pathogens often can be detected in 1 to 2 days.
10. Dairy products, grains, meats, fruits, and vegetables can be fermented. It has been suggested that some fermented dairy products have antimicrobial and anticancer characteristics, especially when particular lactic acid bacteria are used.
11. Wines are produced by the direct fermentation of fruit juices or musts. For fermentation of cereals and grains, starches and proteins contained in these substrates must first be hydrolyzed to provide substrates for the alcoholic fermentation.
12. The making of bread, sauerkraut, sufu, pickles, and many other foods also involves the use of complex fermentation processes. When chopped plant materials are fermented, silages are created, which can be stored and used by animals.
13. Microbial cells can be used as food sources and food amendments. These include mushrooms, cyanobacteria such as *Spirulina*, and yeasts. There is an increasing interest in probiotics, or the use of microorganisms to change the microbial community in the intestine. Microbial colonization of surfaces in the intestine plays a critical role in these processes.

Tell me what you eat, and I will tell you what you are.

—Brillat-Savarin

Foods, microorganisms, and humans have had a long and interesting association that developed long before the beginning of recorded history. Foods are not only of nutritional value to those who consume them but often are ideal culture media for microbial growth. Fermentation by some microorganisms can lead to food preservation instead of food spoilage.

Microorganisms can be used to transform raw foods into gastronomic delights, including cheeses, pickles, sausages, and soy

sauce. Wines, beers, and other alcoholic products also are produced through microbial activity. On the other hand, foods also can serve as vehicles for disease transmission, and the detection and control of pathogens and food spoilage microorganisms are important parts of food microbiology. During the entire sequence of food handling, from the producer to the final consumer, microorganisms can affect food quality and human health.

Wholesome, nutrient-rich foods are important for all people. Microbial growth in foods can result in either preservation or spoilage, depending on the microorganisms involved and the food storage conditions. Contamination by disease-causing microorganisms can occur at any point in the food-handling sequence.

41.1 Microorganism Growth in Foods

Foods, because they provide nutrients for us, also are excellent environments for the growth of microorganisms. Microbial growth is controlled by factors related to the food itself, or **intrinsic factors**, and also to the environment where the food is being stored, or what are described as **extrinsic factors**, as shown in **figure 41.1**.

The intrinsic or food-related factors include pH, moisture content, water activity or availability, oxidation-reduction potential, physical structure of the food, available nutrients, and the possible presence of natural antimicrobial agents. Extrinsic or environmental factors include temperature, relative humidity, gases (CO_2 , O_2) present, and the types and numbers of microorganisms present in the food.

Intrinsic Factors

Food composition is a critical intrinsic factor that influences microbial growth. If a food consists primarily of carbohydrates, spoilage does not result in major odors. Thus foods such as breads, jams, and some fruits first show spoilage by fungal growth. In contrast, when foods contain large amounts of proteins and/or fats (for example, meat and butter), spoilage can produce a variety of foul odors. One only need think of rotting eggs. This proteolysis and anaerobic breakdown of proteins that yields foul-

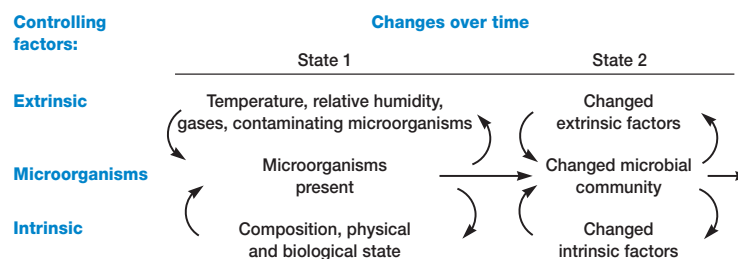


Figure 41.1 Intrinsic and Extrinsic Factors. A variety of intrinsic and extrinsic factors can influence microbial growth in foods. Time-related successional changes occur in the microbial community and the food.

Table 41.1 Differences in Spoilage Processes in Relation to Food Characteristics

Substrate	Food Example	Chemical Reactions or Processes ^a	Typical Products and Effects
Pectin	Fruits	Pectinolysis	Methanol, uronic acids (loss of fruit structure, soft rots)
Proteins	Meat	Proteolysis, deamination	Amino acids, peptides, amines, H ₂ S, ammonia, indole (bitterness, souring, bad odor, sliminess)
Carbohydrates	Starchy foods	Hydrolysis, fermentations	Organic acids, CO ₂ , mixed alcohols (souring, acidification)
Lipids	Butter	Hydrolysis, fatty acid degradation	Glycerol and mixed fatty acids (rancidity, bitterness)

^aOther reactions also occur during the spoilage of these substrates.

smelling amine compounds is called **putrefaction**. One major source of odor is the organic amine cadaverine (imagine the origin of that name). Degradation of fats ruins food as well. For example, the production of short-chained fatty acids from fats renders butter rancid and unpleasant.

The pH of a food also is critical because a low pH favors the growth of yeasts and molds (*see section 6.4*). In neutral or alkaline pH foods, such as meats, bacteria are more dominant in spoilage and putrefaction. Depending on the major substrate present in a food, different types of spoilage may occur (**table 41.1**).

The presence and availability of water also affect the ability of microorganisms to colonize foods. Simply by drying a food, one can control or eliminate spoilage processes. Water, even if present, can be made less available by adding solutes such as sugar and salt. Water availability is measured in terms of water activity (a_w). This represents the ratio of relative humidity of the air over a test solution compared with that of distilled water. When large quantities of salt or sugar are added to food, most microorganisms are dehydrated by the hypertonic conditions and cannot grow (**table 41.2**; *see also table 6.4*). Even under these adverse conditions, osmophilic and xerophilic microorganisms may spoil food. **Osmophilic** [Greek *osmos*, impulse, and *philein*, to love] **microorganisms** grow best in or on media with a high osmotic concentration, whereas **xerophilic** [Greek *xerosis*, dry, and *philein*, to love] **microorganisms** prefer a low a_w environment and may not grow under high a_w conditions. [Water activity and microbial growth \(pp. 121–23\)](#)

The oxidation-reduction potential of a food also influences spoilage. When meat products, especially broths, are cooked, they often have lower oxidation-reduction potentials. These products with their readily available amino acids, peptides, and growth factors are ideal media for the growth of anaerobes, including *Clostridium* (*see table 39.5*).

The physical structure of a food also can affect the course and extent of spoilage. The grinding and mixing of foods such as sausage and hamburger not only increase the food surface area and alter cellular structure, but also distribute contaminating microorganisms throughout the food. This can result in rapid spoilage if such foods are stored improperly. Vegetables and fruits have outer skins (peels and rinds) that protect them from spoilage. Often spoilage microorganisms have specialized enzymes that help them weaken and penetrate protective peels and rinds, especially after the fruits and vegetables have been bruised.

Many foods contain natural antimicrobial substances, including complex chemical inhibitors and enzymes. Coumarins found in fruits and vegetables exhibit antimicrobial activity. Cow's milk and eggs also contain antimicrobial substances. Eggs are rich in the enzyme lysozyme that can lyse the cell walls of contaminating gram-positive bacteria (*see figure 31.9*). Other interesting foods with antimicrobial activities include the hot sauces used on raw oysters and other seafoods. Tabasco and other hot red pepper sauces apparently have particularly desirable antimicrobial characteristics.

Herbs and spices often possess significant antimicrobial substances; generally fungi are more sensitive than most bacteria. Sage and rosemary are two of the most antimicrobial spices. Aldehydic and phenolic compounds are found in cinnamon, mustard, and oregano. These compounds inhibit microbial growth. Other important inhibitors are garlic, which contains allicin, and cloves, which have eugenol. However, spices also can contain pathogenic and spoilage organisms. Coliforms, *B. cereus*, *C. perfringens*, and *Salmonella* species have been detected in most spices. Microorganisms can be eliminated or reduced by ethylene oxide sterilization. This treatment can result in *Salmonella*-free spices and herbs and a 90% reduction in the levels of general spoilage organisms.

Unfermented green and black teas also have well-documented antimicrobial properties because of their polyphenol contents, which apparently are diminished when the teas are fermented. Such teas are active against bacteria, viruses, and fungi and may have anticancer properties.

Extrinsic Factors

Temperature and relative humidity are important extrinsic factors in determining whether a food will spoil. At higher relative humidities microbial growth is initiated more rapidly, even at lower temperatures (especially when refrigerators are not maintained in a defrosted state). When drier foods are placed in moist environments, moisture absorption can occur on the food surface, eventually allowing microbial growth.

The atmosphere in which the food is stored also is important. This is especially true with shrink-packed foods because many plastic films allow oxygen diffusion, which results in increased growth of surface-associated microorganisms. Excess CO₂ can decrease the solution pH, inhibiting microbial growth. Storing

Table 41.2 Approximate Minimum Water Activity Relationships of Microbial Groups and Specific Organisms of Important in Food Spoilage

Organisms	a_w	Organisms	a_w
Groups		Groups	
Moist spoilage bacteria	0.9	Halophilic bacteria	0.75
Most spoilage yeasts	0.88	Xerophilic molds	0.61
Most spoilage molds	0.80	Osmophilic yeasts	0.61
Specific Microorganisms		Specific Microorganisms	
<i>Clostridium botulinum</i> , type E	0.97	<i>Candida scottii</i>	0.92
<i>Pseudomonas</i> spp.	0.97	<i>Trichosporon pullulans</i>	0.91
<i>Acinetobacter</i> spp.	0.96	<i>Candida zeylanoides</i>	0.90
<i>Escherichia coli</i>	0.96	<i>Geotrichum candidum</i>	~ 0.90
<i>Enterobacter aerogenes</i>	0.95	<i>Trichothecium</i> spp.	~ 0.90
<i>Bacillus subtilis</i>	0.95	<i>Byssoschlamys nivea</i>	~ 0.87
<i>Clostridium botulinum</i> , types A and B	0.94	<i>Staphylococcus aureus</i>	0.86
<i>Candida utilis</i>	0.94	<i>Alternaria citri</i>	0.84
<i>Vibrio parahaemolyticus</i>	0.94	<i>Penicillium patulum</i>	0.81
<i>Botrytis cinerea</i>	0.93	<i>Eurotium repens</i>	0.72
<i>Rhizopus stolonifer</i>	0.93	<i>Aspergillus conicus</i>	0.70
<i>Mucor spinosus</i>	0.93	<i>Aspergillus echinulatus</i>	0.64
		<i>Zygosaccharomyces rouxii</i>	0.62
		<i>Xeromyces bisporus</i>	0.51

Adapted from James M. Jay, 2000. *Modern Food Microbiology*, 6th edition. Reprinted by permission of Aspen Publishers, Inc. Gaithersburg, MD. Tables 3–5, p. 42.

meat in a high CO₂ atmosphere inhibits gram-negative bacteria, resulting in a population dominated by the lactobacilli.

The observation that food storage atmosphere is important has led to the development **modified atmosphere packaging (MAP)**. By the use of modern shrink-wrap materials and vacuum technology, it is possible to package foods with controlled atmospheres. With a carbon dioxide content of 60% or greater in the atmosphere surrounding a food, spoilage fungi will not grow, even if low levels of oxygen are present. Some oxygen is kept because if all the oxygen is removed, the psychrophile *Clostridium gasigenes* can grow. This organism can produce gases in 14 days at 2°C, which leads to swollen food packages. Such MAP procedures, while assisting in controlling spoilage, also cause a shift in the general structure of the microbial community, from gram-negative to gram-positive organisms.

1. What are the effects of food composition on spoilage processes?
2. What are some intrinsic factors that influence food spoilage and how do they exert their effects?
3. Why might sausage and other ground meat products provide a better environment for the growth of food spoilage organisms than raw cuts of meats?
4. List some antimicrobial substances found in foods. What is the mechanism of action of lysozyme?
5. What primary extrinsic factors can determine whether food spoilage will occur?
6. What are the major gases involved in MAP? How are their concentrations varied to inhibit microbial growth?

41.2 Microbial Growth and Food Spoilage

Because foods are such excellent sources of nutrients, if the intrinsic and extrinsic conditions are appropriate, microorganisms grow rapidly and make what once was an attractive and appealing food into a sour, foul-smelling or fungus-covered mass suitable only for the garbage can. Microbial growth in foods can lead to visible changes, including a variety of colors caused by spoilage organisms, which often have been associated with “miracles” and “witchcraft.” One of the most famous is the report of “blood” on communion wafers and other bread, called the “Miracle of Bolsena,” which occurred in 1263. The riddle was eventually solved by Bartolomeo Bizio in 1879, when he described the bacterium responsible for this phenomenon. He also named the bacterium *Serratia marcescens*.

Meat and dairy products, with their high nutritional value and the presence of easily usable carbohydrates, fats, and proteins, provide ideal environments for microbial spoilage. Proteolysis and putrefaction are typical results of microbial spoilage of such high-protein materials. Unpasteurized milk undergoes a predictable four-step succession during spoilage; acid production by *Lactococcus lactis* subsp. *lactis* is followed by additional acid production associated with the growth of more acid tolerant organisms such as *Lactobacillus*. At this point yeasts and molds become dominant and degrade the accumulated lactic acid, and the acidity gradually decreases. Eventually protein-digesting bacteria become active, resulting in a putrid odor and bitter flavor. The milk, originally opaque, can eventually become clear (figure 41.2).



Figure 41.2 Spoilage of a Dairy Product. Fresh (left) and curdled (right) milk are shown. The curdled milk has undergone a natural four-step sequence of spoilage organism activity, resulting in separated curds and whey.

In comparison with meat and dairy products, most fruits and vegetables have a much lower protein and fat content and undergo a different kind of spoilage. Readily degradable carbohydrates favor vegetable spoilage by bacteria, especially bacteria that cause soft rots, such as *Erwinia carotovora*, which produces hydrolytic enzymes. The high oxidation-reduction potential and lack of reduced conditions permits aerobes and facultative anaerobes to contribute to the decomposition processes. Bacteria do not seem important in the initial spoilage of whole fruits; instead such spoilage often is initiated by molds. These organisms have enzymes that contribute to the weakening and penetration of the protective outer skin.

Food spoilage problems occur with minimally processed, concentrated frozen citrus products. These are prepared with little or no heat treatment, and major spoilage can be caused by *Lactobacillus* and *Leuconostoc* spp., which produce diacetyl-butter flavors. *Saccharomyces* and *Candida* can also spoil juices. Concentrated juice has a decreased water activity ($a_w = 0.8$ to 0.83), and when kept frozen at about -9°C , the juices can be stored for long periods. However, when concentrated juices are diluted with water that contains spoilage organisms, or if the juice is stored in improperly washed containers, problems can occur. Also, microorganisms in the frozen concentrated juices can begin the spoilage process after addition of water. Ready-to-serve (RTS) juices present other problems as the a_w values are sufficiently high to allow microbial growth. This is especially true with extended storage at refrigeration temperatures. Although pasteurization can be used, most consumers are sensitive to the loss of flavor that this process entails.

Molds are a special problem for tomatoes. Even the slightest bruising of the tomato skin, exposing the interior, will result in rapid



(a)



(b)

Figure 41.3 Food Spoilage. When foods are not stored properly, microorganisms can cause spoilage. Typical examples are fungal spoilage of (a) bread and (b) corn. Such spoilage of corn is called ear rot. This can result in major economic losses.

fungal growth. Frequently observed genera include *Alternaria*, *Cladosporium*, *Fusarium*, and *Stemphylium*. This growth affects the quality of tomato products, including tomato juices and ketchups.

Molds can rapidly grow on grains and corn when these products are held under moist conditions (**figure 41.3**). Infection of grains by the ascomycete *Claviceps purpure* causes **ergotism**, a toxic condition. Hallucinogenic alkaloids produced by this fungus can lead to altered behavior, abortion, and death if infected grains are eaten. Ergotism is further discussed in chapter 25 (*see p. 561*).

Fungus-derived carcinogens include the aflatoxins and fumonisins. **Aflatoxins** are produced most commonly in moist grains and nut products. Aflatoxins were discovered in 1960, when 100,000 turkey poults died from eating fungus-infested peanut meal. *Aspergillus flavus* was found in the infected peanut meal, together with alcohol-extractable toxins termed aflatoxins. These flat-ringed planar compounds intercalate with the cells' nucleic acids and act as frameshift mutagens and carcinogens. This occurs primarily in the liver, where they are converted to unstable derivatives.

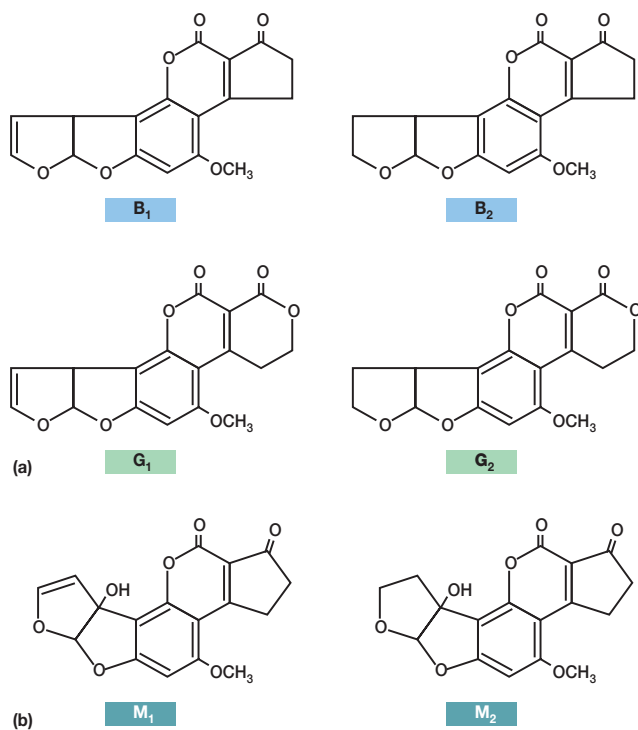


Figure 41.4 Aflatoxins. When *Aspergillus flavus* and related fungi grow on foods, carcinogenic aflatoxins can be formed. These have four basic structures. (a) The letter designations refer to the color of the compounds under ultraviolet light after extraction from the grain and separation by chromatography. The B₁ and B₂ compounds fluoresce with a blue color, and the G₁ and G₂ appear green. (b) The two type M aflatoxins are found in the milk of lactating animals that have ingested type B aflatoxins.

Aflatoxins B₁ and B₂, after ingestion by lactating animals, will be modified in the animal body to yield the aflatoxins M₁ and M₂, noted in **figure 41.4**. If cattle consume aflatoxin-contaminated feeds, these also can appear in milk and dairy products. With the ability to screen grains and corns rapidly for these compounds, it is possible to limit shipments of potentially contaminated grains and feedstuffs. The aflatoxins are potent hepatocarcinogens, which have been linked to effects on immunocompetence, growth, and disease resistance in livestock and laboratory animals. The major aflatoxin types and their derivatives can be separated by chromatographic procedures and can be recognized under UV light by their characteristic fluorescence. Besides their importance in grains, they have also been observed in beer, cocoa, raisins, and soybean meal.

Ultimately, the critical concern is the amounts of aflatoxins that are ingested. Diet preferences appear to be related to aflatoxin exposure: the average aflatoxin intake in the typical European-style diet is 19 ng/day, whereas for Far Eastern diets it is estimated to be 103 ng/day. Aflatoxin sensitivity also can be

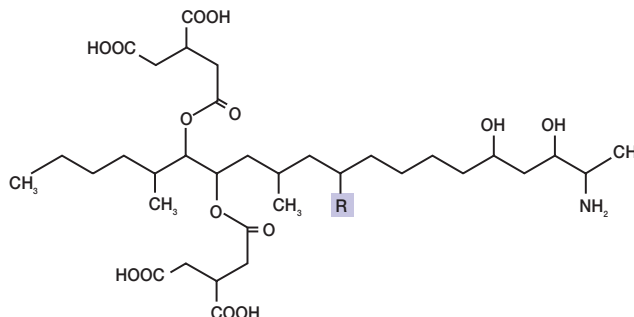


Figure 41.5 Fumonisin Structure. The basic structure of fumonisins FB1 and FB2 produced by *Fusarium moniliforme*, a fungal contaminant that can grow in improperly stored corn. A total of at least ten different fumonisins have been isolated. These are strongly polar compounds that cause diseases in domestic animals and also in humans. FB1, R = -OH; FB2, R = H.

influenced by prior disease exposure. It has been found that individuals who have had hepatitis B have a 30-fold higher risk of liver cancer upon exposure to aflatoxins than individuals who have not had this disease. Based on this linkage, it has been observed that prevention of hepatitis B infections by vaccination and reduction of carrier populations will help to significantly control the potential effects of aflatoxins in foodstuffs.

More recently discovered fungal contaminants of corn are the **fumonisins**, first isolated in 1988. These are produced by *Fusarium moniliforme* and cause leukoencephalomalacia in horses, pulmonary edema in pigs, and esophageal cancer in humans. The fumonisins function by disrupting the synthesis and metabolism of sphingolipids, important biochemically active compounds, which influence a wide variety of cell functions.

There are at least ten different fumonisins; the basic structures of fumonisins FB1 and FB2 are shown in **figure 41.5**. Corn and corn-based feeds and foods, including cornmeal and corn grits, often are contaminated. The fumonisins inhibit ceramide synthase, a key enzyme for the proper use of fatty substances in the cell. Thus it is extremely important to store corn and corn products under dry conditions where these fungi cannot develop.

Eucaryotic microorganisms can synthesize potent toxins other than aflatoxins and fumonisins. **Algal toxins** contaminate fish and thus affect the health of marine animals higher in the food chain; they also can contaminate shellfish and fin fish, which are later consumed by humans. Most toxins are produced by dinoflagellates, but some diatoms also are toxic. Major human diseases that result from algal toxins in marine products include amnesic, diarrhetic, and neurotoxic shellfish poisoning (**table 41.3**). These complex algal toxins (**figure 41.6**), most of which are temperature stable, are known to cause peripheral neurological system effects, often in less than one hour after ingestion. **Toxic algal blooms** (Box 26.2)

Table 41.3 Toxic Syndromes Associated with Marine Algal Toxins

Syndrome	Causative Organism(s)	Primary Vector	Toxin Type
Parasitic shellfish poisoning	<i>Alexandrium</i> spp. <i>Gymnodinium</i> spp. <i>Pyrodinium</i> spp.	Shellfish	Saxitoxins
Neurotoxic shellfish poisoning	<i>Gymnodinium breve</i>	Shellfish	Brevetoxins
Ciguatera fish poisoning	<i>Gambierdiscus toxicus</i>	Reef fish	Ciguatoxins
Amnesic shellfish poisoning	<i>Pseudo-nitzschia</i> spp.	Shellfish	Domoic acid
Diarrhetic shellfish poisoning	<i>Dinophysis</i> spp. <i>Prorocentrum</i> spp.	Shellfish	Dinophysistoxins Okadaic acid
Estuary syndrome	<i>Pfiesteria piscicida</i>	Water	Unknown

Source: F. M. van Dolah, 2000. Marine algal toxins: Origins, health effects, and their increased occurrence. *Environ. Health Perspect.* 108(Suppl. 1):133–141. Table 1, p. 134.

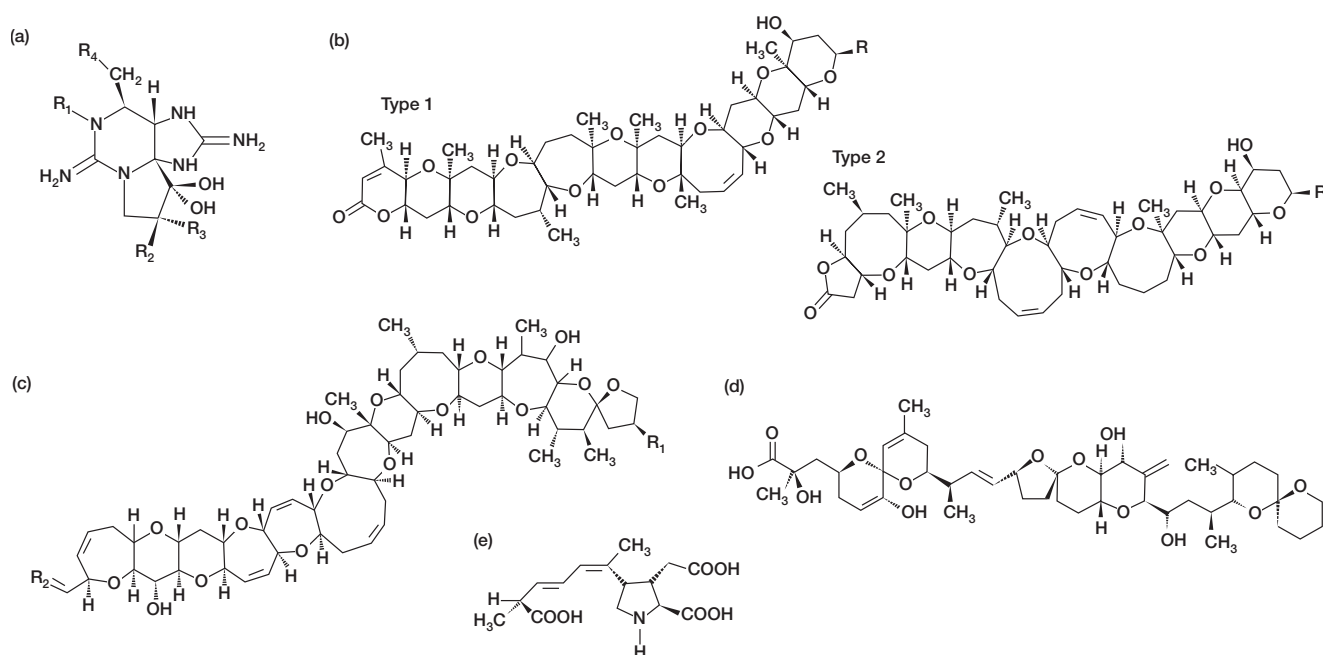


Figure 41.6 Marine Algal Toxins. Structures of marine algal toxins that can impact humans through seafoods and water: (a) saxitoxin, (b) brevetoxin, (c) ciguatoxin, (d) okadaic acid, (e) domoic acid.

1. What fungal genus produces ergot alkaloids? What conditions are required for the synthesis of these substances?
2. Aflatoxins are produced by which fungal genus? How do they damage animals that eat the contaminated food?
3. What microbial genus produces fumonisins and why are these compounds of concern? If improperly stored, what are the major foods and feeds in which these chemicals might be found?
4. Describe in general how food spoilage occurs. What factors influence the nature of the spoilage organisms responsible?
5. Why do concentrated citrus juices present such interesting spoilage problems?
6. What is the usual route by which humans consume algal toxins? What are the major groups of algae that produce these complex substances?

41.3 Controlling Food Spoilage

With the beginning of agriculture and a decreasing dependence on hunting and gathering, the need to preserve surplus foods became essential to survival. The use of salt as a meat preservative and the production of cheeses and curdled milks was introduced in Near Eastern civilization as early as 3000 B.C. The production of wines and the preservation of fish and meat by smoking also were common by this time. Despite a long tradition of efforts to preserve food from spoilage, it was not until the nineteenth century that the microbial spoilage of food was studied rigorously. Louis Pasteur established the modern era of food microbiology in 1857, when he showed that microorganisms cause milk spoilage. Pasteur's work in the 1860s proved that heat could be used to control spoilage organisms in wines and beers (*see section 1.4*).

Control of microorganism growth (chapter 7)

Foods can be preserved by a variety of methods (**table 41.4**). It is vital to eliminate or reduce the populations of spoilage and disease-causing microorganisms and to maintain the microbiological quality of a food with proper storage and packaging. Contamination often occurs after a package or can is opened and just before the food is served. This can provide an ideal opportunity for growth and transmission of pathogens, if care is not taken.

Removal of Microorganisms

Microorganisms can be removed from water, wine, beer, juices, soft drinks, and other liquids by filtration. This can keep bacterial populations low or eliminate them entirely. Prefilters and centrifugation often are used to maximize filter life and effectiveness. Several major brands of beer are filtered rather than pasteurized to better preserve the flavor and aroma of the original product.

Low Temperature

Refrigeration at 5°C retards microbial growth, although with extended storage, psychrophiles and psychrotrophs will eventually grow and produce spoilage. Slow microbial growth at tempera-

Table 41.4 Basic Approaches to Food Preservation

Approach	Examples of Process
Removal of microorganisms	Avoidance of microbial contamination; physical filtration, centrifugation
Low temperature	Refrigeration, freezing
High temperature	Partial or complete heat inactivation of microorganisms (pasteurization and canning)
Reduced water availability	Water removal, as with lyophilization or freeze drying; use of spray dryers or heating drums; decreasing water availability by addition of solutes such as salt or sugar
Chemical-based preservation	Addition of specific inhibitory compounds (e.g., organic acids, nitrates, sulfur dioxide)
Radiation	Use of ionizing (gamma rays) and nonionizing (UV) radiation
Microbial product-based inhibition	The addition of substances such as bacteriocins to foods to control food-borne pathogens

tures below -10°C has been described, particularly with fruit juice concentrates, ice cream, and some fruits. Some microorganisms are very sensitive to cold and their numbers will be reduced, but cold does not lead to significant decreases in overall microbial populations. [Temperature effects on microbial growth \(pp. 125–27\)](#)

High Temperature

Controlling microbial populations in foods by means of high temperatures can significantly limit disease transmission and spoilage.

Heating processes, first used by Nicholas Appert in 1809 (**Box 41.1**), provide a safe means of preserving foods, particularly when carried out in commercial **canning** operations (**figure 41.7**). Canned food is heated in special containers called retorts at about 115°C for intervals ranging from 25 to over 100 minutes. The precise time and temperature depend on the nature of the food. Sometimes canning does not kill all the microorganisms, but only those that will spoil the food (remaining bacteria are unable to grow due to acidity of the food, as an example). After heat treatment the cans are cooled as rapidly as possible, usually with cold water. Quality control and processing effectiveness are sometimes compromised, however, in home processing of foods, especially with less acidic (pH values greater than 4.6) products such as green beans or meats. [Destruction of microorganisms by heat \(pp. 139–42\)](#)

Pasteurization involves heating food to a temperature that kills disease-causing microorganisms and substantially reduces the levels of spoilage organisms. In the processing of milk, beers, and fruit juices by conventional low-temperature holding (LTH) pasteurization, the liquid is maintained at 62.8°C for 30 minutes. Products can also be held at 71°C for 15 seconds, a high-temperature, short-time (HTST) process; milk can be treated at 141°C for 2 seconds for ultra-high-temperature (UHT) processing. Shorter-term processing results in improved flavor and extended product shelf life.

Such heat treatment is based on a statistical probability that the number of remaining viable microorganisms will be below a

Box 41.1

An Army Travels on Its Stomach

The movement and maintenance of large numbers of military personnel have always been limited by food supplies. The need to maintain large numbers of troops under hostile and inclement conditions led the French government in 1795 to offer a prize of 12,000 francs to the individual who could preserve foods for use under field conditions. Eventually the prize was awarded to Nicholas Appert, a candy maker, for his development of a heating process in which meats and other products could be preserved under sealed conditions.

Appert's work was based on the assumptions that heating and boiling control "ferments" and that sealing the food in bottles before heating it avoids the effects of air on spoilage. Despite Leeuwenhoek's earlier work, Appert did not have the concept of microorganisms to assist him in explaining the effectiveness of his process. His containers were large glass bottles, sealed with laminated corks and fish glue. With extreme care and attention to detail, he was able to heat these bottles in boiling water to provide food that could be stored for several years. Appert's work was an important foundation for the later studies of Louis Pasteur.



Figure 41.7 A Canning Operation. Microbial control is important in the processing and preservation of many foods. Worker pouring peas into a large, clean vat during the preparation of vegetable soup. After preparation the soup is transferred to cans. Each can is heated for a short period, sealed, processed at temperatures around 110–121°C in a canning retort to destroy spoilage microorganisms, and finally cooled.



Figure 41.8 Food Preservation by Canning. This technique is widely used and very effective. Improper canning can occasionally occur, as shown by this bent and leaking can.

certain level after a particular heating time at a specific temperature. This process is discussed in detail on pp. 138–40.

Despite efforts to eliminate spoilage microorganisms during canning, sometimes canned foods are spoiled (**figure 41.8**). This may be due to spoilage before canning, underprocessing during canning, and leakage of contaminated water through can seams during cooling. Spoiled food can be altered in such characteristics as color, texture, odor, and taste. Organic acids, sulfides, and gases (particularly CO₂ and H₂S) may be produced. In flat sour spoilage no gas is generated and the can does not swell, but its contents are rendered sour by the presence of fermentation acids. If spoilage microorganisms produce gas, both ends of the can will bulge outward to give a swell. Sometimes the swollen ends can be moved by thumb pressure (soft swells); in other cases the gas pressure is so great that the ends cannot be dented by hand (hard swells). It should be noted that swelling is not always due to microbial spoilage. Acid in high-acid foods may react with the iron of the can to release hy-

drogen and generate a hydrogen swell. Hydrogen sulfide production by *Desulfotomaculum* can cause "sulfur stinkers."

Water Availability

Dehydration, such as lyophilization to produce freeze-dried foods, is now a common means of eliminating microbial growth. The modern process is simply an update of older procedures in which grains, meats, fish, and fruits were dried. The combination of free-water loss with an increase in solute concentration in the remaining water makes this type of preservation possible.

Chemical-Based Preservation

Various chemical agents can be used to preserve foods, and these substances are closely regulated by the U.S. Food and Drug Administration and are listed as being "generally recognized as safe" or

Table 41.5 Major Groups of Chemicals Used in Food Preservation

Preservatives	Approximate Maximum Use Range	Organisms Affected	Foods
Propionic acid/propionates	0.32%	Molds	Bread, cakes, some cheeses, inhibitor of rropy bread dough
Sorbic acid/sorbates	0.2%	Molds	Hard cheeses, figs, syrups, salad dressings, jellies, cakes
Benzoic acid/benzoates	0.1%	Yeasts and molds	Margarine, pickle relishes, apple cider, soft drinks, tomato ketchup, salad dressings
Parabens ^a	0.1%	Yeasts and molds	Bakery products, soft drinks, pickles, salad dressings
SO ₂ /sulfites	200–300 ppm	Insects and microorganisms	Molasses, dried fruits, wine, lemon juice (not to be used in meats or other foods recognized as sources of thiamine)
Ethylene/propylene oxides	700 ppm	Yeasts, molds, vermin	Fumigant for spices, nuts
Sodium diacetate	0.32%	Molds	Bread
Dehydroacetic acid	65 ppm	Insects	Pesticide on strawberries, squash
Sodium nitrite	120 ppm	Clostridia	Meat-curing preparations
Caprylic acid	—	Molds	Cheese wraps
Ethyl formate	15–200 ppm	Yeasts and molds	Dried fruits, nuts

From James M. Jay. 2000. *Modern Food Microbiology*, 6th edition. Reprinted by permission of Aspen Publishing, Frederick, Md.

^aMethyl-, propyl-, and heptyl-esters of *p*-hydroxybenzoic acid.

GRAS (table 41.5). They include simple organic acids, sulfite, ethylene oxide as a gas sterilant, sodium nitrite, and ethyl formate. These chemical agents affect microorganisms by disrupting a critical cell factor. For example, they may damage the plasma membrane or denature various cell proteins. Still other compounds interfere with the functioning of nucleic acids, thus inhibiting cell reproduction.

The effectiveness of many of these chemical preservatives depends on the food pH. As an example, sodium propionate is most effective at lower pH values, where it is primarily undissociated and able to be taken up by lipids of microorganisms. Breads, with their low pH values, often contain sodium propionate as a preservative. Chemical preservatives are used with grain, dairy, vegetable, and fruit products. Sodium nitrite is an important chemical used to help preserve ham, sausage, bacon, and other cured meats by inhibiting the growth of *Clostridium botulinum* and the germination of its spores. This protects against botulism and reduces the rate of spoilage. Besides increasing meat safety, nitrite decomposes to nitric acid, which reacts with heme pigments to keep the meat red in color. Current concern about nitrite arises from the observation that it can react with amines to form carcinogenic nitrosamines. Nitrite is added in very small amounts, and eventually it may be possible to eliminate its use entirely.

Radiation

Radiation, both ionizing and nonionizing, has an interesting history in relation to food preservation. Ultraviolet radiation is used to control populations of microorganisms on the surfaces of laboratory and food-handling equipment, but it does not penetrate food. The major method used for radiation sterilization of food is

gamma irradiation from a cobalt-60 source (*see p. 144*). Such electromagnetic radiation has excellent penetrating power and must be used with moist foods because the radiation produces peroxides from water in the microbial cells, resulting in oxidation of sensitive cellular constituents. This process of **radappertization**, named after Nicholas Appert, can extend the shelf life of seafoods, fruits, and vegetables. To sterilize meat products, commonly 4.5 to 5.6 megarads are used.

Among the more interesting radiation-resistant bacteria that have been studied is *Deinococcus radiodurans* (*see section 21.2*). This bacterium has a complex cell wall structure and tetrad-forming growth patterns (*see figure 21.2*). It also has an extraordinary capacity to withstand high doses of radiation, although the mechanism for its resistance is not understood.

Microbial Product–Based Inhibition

There is increasing interest in the use of **bacteriocins** (*see p. 712*) for the preservation of foods. Bacteriocins are bactericidal proteins active against closely related bacteria, which bind to specific sites on the cell, and affect cell membrane integrity and function. The only currently approved product is nisin. Nisin, produced by some strains of *Streptococcus lactis*, is a small hydrophobic protein. It is nontoxic to humans and affects mainly gram-positive bacteria, especially *Enterococcus faecalis*. Nisin can be used particularly in low-acid foods to improve inactivation of *Clostridium botulinum* during the canning process or to inhibit germination of any surviving spores.

Bacteriocins function by dissipating the proton motive force (PMF) of a susceptible bacterium. These compounds have a wide

Table 41.6 Major Food-Borne Infectious Diseases

Disease	Organism	Incubation Period and Characteristics	Major Foods Involved
Salmonellosis	<i>S. typhimurium</i> , <i>S. enteritidis</i>	8–48 hr Enterotoxin and cytotoxins	Meats, poultry, fish, eggs, dairy products
<i>Arcobacter</i> diarrhea	<i>Arcobacter butzleri</i>	Severe diarrhea, recurrent cramps	Meat products, especially poultry
Campylobacteriosis	<i>Campylobacter jejuni</i>	Usually 2–10 days Most toxins are heat-labile	Milk, pork, poultry products, water
Listeriosis	<i>L. monocytogenes</i>	Varying periods Related to meningitis and abortion; newborns and the elderly especially susceptible	Meat products, especially pork and milk
<i>Escherichia coli</i> diarrhea and colitis	<i>E. coli</i> , including serotype O157:H7	24–72 hr Enterotoxigenic positive and negative strains; hemorrhagic colitis	Undercooked ground beef, raw milk
Shigellosis	<i>Shigella sonnei</i> , <i>S. flexneri</i>	24–72 hr	Egg products, puddings
Yersiniosis	<i>Yersinia enterocolitica</i>	16–48 hr Some heat-stable toxins	Milk, meat products, tofu
<i>Plesiomonas</i> diarrhea	<i>Plesiomonas shigelloides</i>	1–2 hr	Uncooked mollusks and foreign travel
<i>Vibrio parahaemolyticus</i> gastroenteritis	<i>V. parahaemolyticus</i>	16–48 hr	Seafood, shellfish

variety of names, depending on the organisms that produce them. Bacteriocins function by the formation of hydrophilic pores in a susceptible bacterium and the release of low-molecular-weight molecules; this may occur as a result of peptidoglycan synthesis inhibition and detergent-like effects on the cytoplasmic membrane. Bacteriocin addition to foods such as cheddar cheese can lead to a two- to threefold reduction in *Listeria monocytogenes* in 180-day-old cheeses. Similar compounds also occur in the eucaryotes.

1. Describe the major approaches used in food preservation.
2. What types of chemicals can be used to preserve foods?
3. Nitrite is often used to improve the storage characteristics of prepared meats. What toxicological problems may result from the use of this chemical?
4. Under what conditions can ultraviolet light and gamma radiation be used to control microbial populations in foods and in food preparation? What is radappertization?
5. In principle, how do bacteriocins such as nisin function? What bacterial genus produces this important polypeptide?

41.4 Food-Borne Diseases

Food-borne illnesses impact the entire world. In the United States, based on recent information from the Centers for Disease Control and Prevention, annual incidences of food-related diseases involve 76 million cases, of which only 14 million can be attributed to known pathogens. Food-borne diseases result in 325,000 hospitalizations and at least 5,000 deaths per year. Since 1942, the number of recognized food-borne pathogens has increased over fivefold! Are these new microorganisms? In most cases, these pathogens are simply agents that we now can describe, based on an improved un-

derstanding of microbial diversity. Recent estimates indicate that Norwalk-like viruses, *Campylobacter jejuni* and *Salmonella* are the major causes of food-borne diseases. In addition, *Escherichia coli* O157:H7 and *Listeria* are important food-related pathogens.

Many diseases transmitted by foods, or food poisonings, are discussed in chapters 38 and 39, and only a few of the more important food-borne bacterial pathogens are mentioned here. There are two primary types of food-related diseases: food-borne infections and food intoxications. [Food-borne and waterborne diseases \(pp. 891–93, 926–33\)](#)

All these food-borne diseases are associated with poor hygienic practices. Whether by water or food transmission, the fecal-oral route is maintained, with the food providing the vital link between hosts. Fomites, such as sink faucets, drinking cups, and cutting boards, also play a role in the maintenance of the fecal-oral route of contamination.

Food-Borne Infection

A **food-borne infection** involves the ingestion of the pathogen, followed by growth in the host, including tissue invasion and/or the release of toxins. The major diseases of this type are summarized in [table 41.6](#) (see also [table 39.5](#)).

Salmonellosis results from ingestion of a variety of *Salmonella* serovars, particularly *typhimurium* and *enteritidis* (see also [section 39.4](#)). Gastroenteritis is the disease of most concern in relation to foods such as meats, poultry, and eggs, and the onset of symptoms occurs after an incubation time as short as 8 hours. *Salmonella* infection can arise from contamination by workers in food-processing plants and restaurants, as well in canning processes (**Box 41.2**).

Campylobacter jejuni is considered a leading cause of acute bacterial gastroenteritis in humans and can affect persons of all ages. This important pathogen is often transmitted by uncooked or poorly cooked poultry products. For example, transmission

Box 41.2

Typhoid Fever and Canned Meat

Minor errors in canning have led to major typhoid outbreaks. In 1964 canned corned beef produced in South America was cooled, after sterilization, with nonchlorinated water; the vacuum created when the cans were cooled drew *S. typhi* into some of the cans, which were not completely sealed. This contaminated product was later sliced in an Aberdeen, Scotland, food store, and the meat slicer became a continuing con-

tamination source; the result was a major epidemic that involved 400 people. The *S. typhi* was a South American strain, and eventually the contamination was traced to the contaminated water used to cool the cans.

This case emphasizes the importance of careful food processing and handling to control the spread of disease during food production and preparation.

often occurs when kitchen utensils and containers are used for chicken preparation and then for salads. Contamination with as few as 10 viable *Campylobacter jejuni* cells can lead to the onset of diarrhea. *Campylobacter jejuni* also is transmitted by raw milk, and the organism has been found on various red meats. Thorough cooking of food prevents this disease transmission problem.

Listeriosis, caused by *Listeria monocytogenes* (see section 39.4), is of continuing interest, as shown by the outbreak that occurred in Southern California in 1985. This outbreak was caused by improper pasteurization of milk used in the commercial production of Mexican-style cheeses. At least 86 cases of infection occurred, including 58 cases involving mother-infant pairs. Forty-seven people died. The outbreak was traced to pinhole leaks in the heat exchangers of a pasteurizing unit. The leaks allowed incoming raw milk to contaminate the pasteurized milk before production of the cheese. *Listeria* is difficult to work with because an extended incubation of samples is required for growth and detection.

Escherichia coli is now recognized as an important food-borne disease organism. Enteropathogenic, enteroinvasive, and enterotoxigenic types can cause diarrhea. *E. coli* O157:H7 (see p. 932) with its specific somatic (O) and flagellar (H) antigens (see Box 33.3), is thought to have acquired enterohemorrhagic genes from *Shigella*, including the genes for shigalike verocytotoxins. This produced a new pathogenic strain, first discovered in 1982 and now known around the world. The pathogen is spread by the fecal-oral route, and an infectious dose appears to be only 500 bacteria. Enterohemorrhagic *E. coli* has been found in meat products such as hamburger and salami, in unpasteurized fruit drinks, on fruits and vegetables, and in untreated well water. In August 1997, 25 million pounds of hamburger were recalled by a meat processing plant because of *E. coli* O157:H7 contamination. A few years ago a child died in Denver, Colorado, after drinking a popular unpasteurized fruit drink that contained the organism. Those most susceptible to this pathogen are the young.

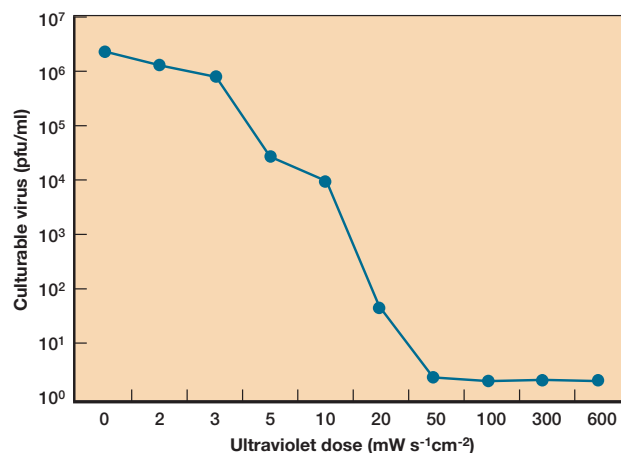
Prevention of food contamination by *E. coli* O157:H7 is essential from the time of production until consumption. Hygiene must be monitored carefully in larger-volume slaughterhouses where contact of meat with fecal material can occur. Even fruits and vegetables should be handled with care because disease outbreaks have been caused by imported produce. Caution also is essential at the point of use. For example, avoidance of food contamination by hands and utensils is critical. Utensils used with raw

foods should not contact cooked food; proper cleaning of cutting boards and utensils minimizes contamination. It may be possible to reduce this threat by destroying the pathogen with gamma irradiation, a food preservation method under consideration for wider use.

An infectious agent of increasing worldwide concern with respect to food safety is a prion that causes the poorly understood new variant Creutzfeldt-Jakob Disease (vCJD). This is one of a group of progressively degenerative neuronal diseases termed transmissible spongiform encephalopathies (TSEs), and is associated with beef cattle. It is often called the “mad cow disease.” With recent occurrences of new variant CJD in humans, bans on beef importation have been put in place in many areas. Major problems in controlling new variant CJD include the long incubation period, which can last many years before onset of the invariably fatal disease, and the lack of reliable detection methods. The major means of vCJD transmission between animals is the use of mammalian tissue in ruminant animal feeds; at the present time, there are major problems in detecting such prohibited animal products in ruminant feeds. [Creutzfeldt-Jakob Disease \(CJD\)](#), [section 38.5](#); [Prions](#), [see section 18.9](#)

Foods that are transported and consumed in an uncooked state are an increasingly important source of food-borne infection. This is especially of concern when there is a possibility of contact with contaminated water at any point between production and consumption. The problem becomes more critical with increasingly rapid movement of people and products around the world. International trade in uncooked foods, aided by rapid air transport, provides many opportunities for disease transmission. Fresh foods such as sprouts, seafood, and raspberries pose significant hazards, which will be discussed here.

Sprouts are an increasingly popular part of the new “healthy” life-style. They are fresh, delicate and form an exciting garnish to complement a variety of foods. Unfortunately, if these are not germinated in pathogen-free waters and grown under sanitary conditions, major growth of pathogens can occur. Care must be taken especially when the seeds are germinated, because a newly germinating seed releases organic matter and creates a **spermosphere** that stimulates microbial growth in a way similar to that which occurs in the rhizosphere. Sprouts produced in areas of the world where there is poor control of water quality and sanitation should be used with caution. Contaminated alfalfa, beans, watercress, mungbean, mustard, and soybean sprouts can be major sources of typhoid and cholera.



RT-PCR	+	+	+	+	+	+	+	+	+
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Figure 41.9 Cultural versus Molecular-Based Virus Detection.

Comparison of plaque-forming ability and recovery of RNA from poliovirus type 2 by RT-PCR with varying UV doses. Even when plaque-forming ability is lost, nucleic acids are still detectable.

Shellfish and finfish also present major concerns. Raw sewage can contaminate shellfish-growing areas; in addition, waterborne pathogens such as *Vibrio* are more prevalent in the water column during the warm months (e.g., in Chesapeake Bay on the mid-Atlantic coast of the United States). Viruses also can be a problem. Oysters are filter feeders that process several liters of water per day, leading to the potential concentration of at least 100 types of enteric viruses. Reverse transcriptase PCR (RT-PCR) can be used to detect RNA viruses in oysters based on the presence of their nucleic acids. However, the inability of molecular techniques to differentiate between infectious and noninfectious particles is a major problem. For example, UV treatment can inactivate many RNA viruses without eliminating the RT-PCR signal, although the virions no longer replicate in a suitable tissue culture environment (**figure 41.9**). Heavy rainfall in shellfish areas can cause runoff of pathogens from adjacent summer cottage septic systems and contaminate coastal waters. Often it is necessary to ban shellfish harvesting until the animals void pathogens from their digestive systems. Alternatively, shellfish from contaminated areas can be moved to clean waters to allow them to clean their digestive systems. [The polymerase chain reaction \(pp. 326–27\)](#)

Raspberries provide an important example of another major problem: the rapid air transport of raw agricultural products around the world. A major recent outbreak of *Cyclospora cayetanensis* poisoning was traced to raspberries imported from Central America into the United States and Canada. In the growth and harvesting process, the raspberries apparently became contaminated, resulting in serious diarrhea in affected individuals. This

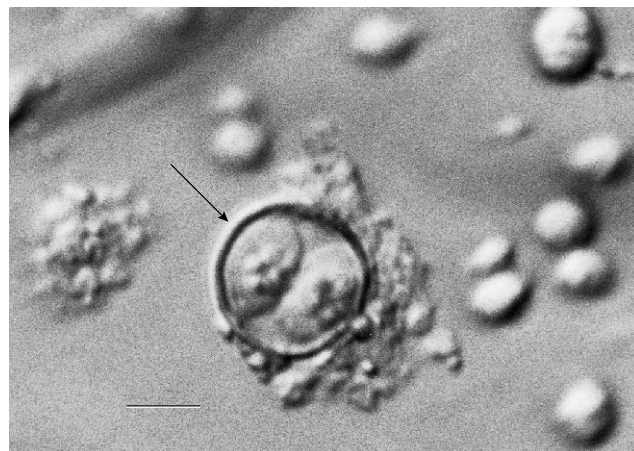


Figure 41.10 *Cyclospora cayetanensis*, an Important Contaminant of Raw Foods. *Cyclospora cayetanensis* can be recognized in waste waters and after recovery from contaminated foods due to the occurrence of an oocyst with two sporocysts. Bar = 5 μ m.

organism has a complex life cycle, which is not fully understood at the present time. In comparison with *Giardia* and *Cryptosporidium*, which are infective immediately after being shed in feces, *Cyclospora* is not immediately infectious; sporulation or maturing requires 12 hours after release from the body. The mature infective cyst has two sporocysts (**figure 41.10**), an important criterion for confirming the presence of this parasite on foods or in the environment. [Cyclospora \(p. 653\)](#)

1. What are the major food-borne diseases in the United States?
2. Discuss the major characteristic of a food-borne infection in terms of the time required between ingestion of the pathogen and the onset of the disease. Why does this occur?
3. What common food is often related to *Campylobacter*-caused gastroenteritis? What means can be used to control the occurrence of this disease from this source?
4. Give some of the sources of *E. coli* O157:H7 that have been of concern in terms of disease transmission.
5. Why is new variant Creutzfeldt-Jakob disease (vCJD) of such concern?
6. What are some uncooked foods that have been implicated in food-borne disease transmission?
7. Why is the spermosphere important for pathogen survival in sprouts?

Food-Borne Intoxications

Microbial growth in food products also can result in a **food intoxication**, as summarized in table 39.5 (see pp. 928–29). Intoxication produces symptoms shortly after the food is consumed because growth of the disease-causing microorganism is not required. Toxins produced in the food can be associated with microbial cells or can be released from the cells.

Most *Staphylococcus aureus* strains cause a staphylococcal enteritis related to the synthesis of extracellular toxins (see section 39.4). These are heat-resistant proteins, and heating will not usually render the food safe. The effects of the toxins are quickly felt, with disease symptoms occurring within 2 to 6 hours. The main reservoir of *S. aureus* is the human nasal cavity. Frequently *S. aureus* is transmitted to a person's hands and then is introduced into food during preparation. Growth and enterotoxin production usually occur when contaminated foods are held at room temperature for several hours.

Three gram-positive rods are known to cause food intoxications: *Clostridium botulinum*, *C. perfringens*, and *Bacillus cereus* (see table 39.5). *C. botulinum* poisoning is discussed in chapter 39, and *C. perfringens* intoxication is described here.

Clostridium perfringens food poisoning is one of the more widespread food intoxications. These microorganisms, which produce exotoxins, must grow to levels of approximately 10^6 bacteria per gram or higher in a food to cause disease. At least 10^8 bacteria must be ingested. They are common inhabitants of soil, water, food, spices, and the intestinal tract. Upon ingestion the cells sporulate in the intestine. The enterotoxin is a spore-specific protein and is produced during the sporulation process. Enterotoxin can be detected in the feces of affected individuals. *Clostridium perfringens* food poisoning is common and occurs after meat products are heated, which results in O_2 depletion. If the foods are cooled slowly, growth of the microorganism can occur. At 45°C , enterotoxin can be detected 3 hours after growth is initiated. Onset of the symptoms—watery diarrhea, nausea, and abdominal cramps—usually occurs in about 8 to 16 hours.

Baked potatoes served in aluminum foil can provide a unique environment for pathogenic microorganisms. Potatoes, even after washing, are covered by *Clostridium botulinum*, which naturally occurs in the soil. If the aluminum foil-covered potatoes are not heated sufficiently in the baking process, surviving clostridia can proliferate after removal of the potatoes from the oven and rapidly produce toxins.

Bacillus cereus also is of concern in starchy foods. It can cause two distinct types of illnesses depending on the type of toxin produced: an emetic illness characterized by nausea and vomiting with an incubation time of 1 to 6 hours, and a diarrheal type, with an incubation of 4 to 16 hours. The emetic type is often associated with boiled or fried rice, while the diarrheal type is associated with a wider range of foods.

1. What are some of the major genera involved in food-borne intoxications?
2. How does a food-borne intoxication differ from a food-borne infection?

41.5 Detection of Food-Borne Pathogens

A major problem in maintaining food safety is the need to rapidly detect microorganisms in order to curb outbreaks that can affect large populations. This is especially important because of widescale

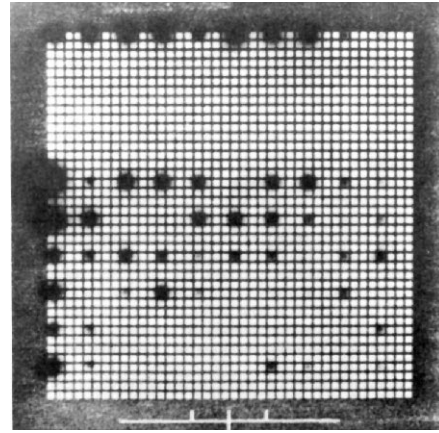


Figure 41.11 Molecular Probes and Food Microbiology.

Molecular techniques are finding increasing use in microbial analysis of foods. Autoradiogram of a radioactive-labeled *Listeria monocytogenes* probe against 100 *Listeria* cultures. Only the *Listeria monocytogenes* cultures show sequence homology and binding with the DNA probe, darkening the autoradiogram film. The other *Listeria* spp. do not react with the probe.

distribution of perishable foods. Standard culture techniques (see chapter 5) may require days to weeks for positive identification of pathogens. Identification is often complicated by the low numbers of pathogens compared with the background microflora. Furthermore, the varied chemical and physical composition of foods can make isolation difficult. Fluorescent antibody, enzyme-linked immunoassays (ELISAs), and radioimmunoassay techniques have proven of value (see section 33.3). These can be used to detect small amounts of pathogen-specific antigens.

Molecular techniques also are increasingly used in identification. The basic methods of analysis and manipulation of DNA and RNA are discussed in chapter 14. These methods are valuable for three purposes: (1) to detect the presence of a single, specific pathogen; (2) to detect viruses that cannot be grown conveniently; and (3) to identify slow-growing or noncultured pathogens.

Pathogens now can be identified by detecting specific DNA or RNA base sequences with **probes**. These usually are 14 to 40 bases in length. They may be created by generating fragments with restriction endonucleases or through direct chemical synthesis. Probes are labeled by linking them to a variety of enzymatic, isotopic, chromogenic, or luminescent/fluorescent markers. A major advantage of their use is the speed with which specific microorganisms can be detected in a set of cultures, as shown in **figure 41.11**. In this example, a hydrophobic grid-membrane system has been used. The *Listeria monocytogenes* cultures are radioactive, indicating that they have bound the probe, while other *Listeria* species do not show probe binding.

Another example of the use of molecular techniques is provided by pathogenic *E. coli*. Currently *E. coli* O157:H7 is isolated and identified using selective culture media, rapid identification kits (see **figure 41.12**), rapid probe-based identification procedures, serotype-specific probes, and PCR techniques. These tech-

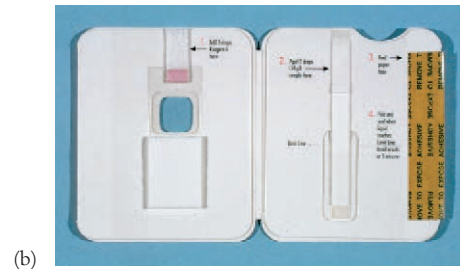


Figure 41.12 A Rapid Test Kit for the Identification of *Escherichia coli* O157:H7 after Growth in an Enrichment Medium. (a) Complete test kit containing the immunochromatographic membrane assay; (b) interior with sample analysis instructions.

niques also have been extended to allow the detection of a few target cells in large populations of background microorganisms. For example, by using the polymerase chain reaction, as few as 10 toxin-producing *E. coli* cells can be detected in a population of 100,000 cells isolated from soft-cheese samples.

As few as two colony forming units of *Salmonella* can be detected with a special PCR system (figure 41.13). This procedure makes it possible to confirm *Salmonella* presence within 24 to 48 hours, whereas 3 to 4 days is needed for presumptive identification with standard culture procedures. Confirmation of *Salmonella* presence would then require additional time. Frequently, to improve the sensitivity and increase the speed of this method, a preenrichment step will be used before PCR. In this type of analysis, especially for well-characterized pathogens, all culture and PCR reagents are provided in “pill” form to speed the entire process. PCR is now being used for rapid detection of food-borne pathogens. In recent studies, the construction of specific 159 and 1,223 base pair PCR products, which can be separated electrophoretically, has made it possible to detect *Campylobacter jejuni* and *Arcobacter butzleri* in the same sample within 8 hours.

A major advance in the detection of food-borne pathogens is the use of standardized pathogen DNA patterns, or “food-borne pathogen fingerprinting.” The Centers for Disease Control and Prevention in the United States has established a new program, called **PulseNet**, in which pulsed-field gel electrophoresis (PFGE) is used under carefully controlled and duplicated conditions to determine the distinctive DNA pattern of each bacterial pathogen. With this uniform procedure, it is possible to link pathogens associated with disease outbreaks in different parts of the world to a specific food source. Data from around the world are being used in **FoodNet**, an active surveillance network, to follow nine major food-borne diseases. Using the FoodNet approach, it is possible to trace the course and cause of infection in days and not weeks. As an example, a *Shigella* outbreak in three different areas of North America was traced to Mexican parsley that had been tainted with polluted irrigation water. This program has resulted in more rapid

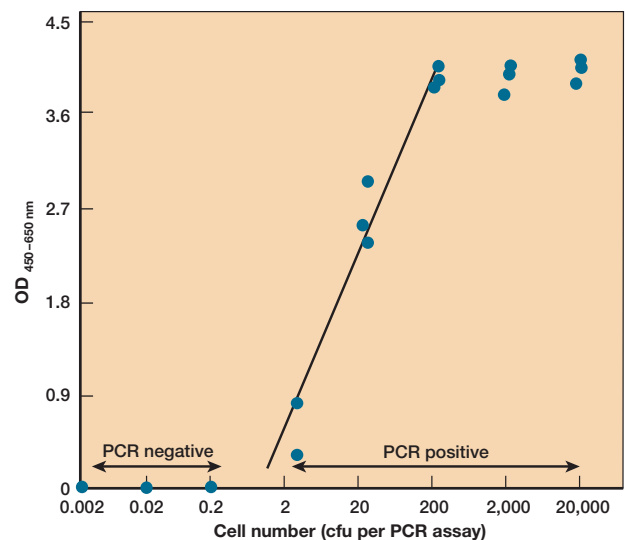


Figure 41.13 Polymerase Chain Reaction (PCR)-Based Pathogen Detection. Comparison of PCR sensitivity and growth for *Salmonella agona* detection. The Probalia™ PCR system can detect as few as two colony forming units (CFU) of the pathogen. OD = optical density.

establishment of epidemiological linkages and a decreased occurrence of many of these important food-borne diseases.

1. What is a “probe” and how is it used for detection of food-borne pathogens?
2. How is the polymerase chain reaction used in pathogen detection?
3. How are PulseNet and FoodNet used in the surveillance of food-borne diseases?

41.6 Microbiology of Fermented Foods

Over the last several thousand years, fermentation has been a major way of preserving food. Microbial growth, either of natural or inoculated populations, causes chemical and/or textural changes to form a product that can be stored for extended periods. The fermentation process also is used to create new, pleasing food flavors and odors.

The major fermentations used in food microbiology are the lactic, propionic, and ethanolic fermentations, as discussed in section 9.3. These fermentations are carried out with a wide range of cultures, many of which have not been characterized.

Fermented Milks

Throughout the world, at least 400 different fermented milks are produced. These fermentations are carried out by mesophilic,

thermophilic, and therapeutic lactic acid bacteria, as well as by yeasts and molds as noted in **table 41.7**. Only major examples of these fermentation types will be discussed in this section.

Mesophilic

Mesophilic milk fermentations result from similar manufacturing techniques, in which acid produced through microbial activity causes protein denaturation. To carry out the process, one usually inoculates milk with the desired starter culture (**Box 41.3**); incubates it at optimum temperature (approximately 20 to 30°C), and then stops microbial growth by cooling. *Lactobacillus* spp. and *Lactococcus lactis* cultures are used for aroma and acid production. The organism *Lactococcus lactis* subsp. *diacetylactis* converts milk citrate to diacetyl, which gives a special buttery flavor to the finished product. The use of these microorganisms with skim milk produces cultured buttermilk, and when cream is used, sour cream is the result. [The genus *Lactobacillus* \(pp. 529–30\)](#); [The genus *Lactococcus* \(pp. 531–33\)](#)

Thermophilic

In addition to mesophilic milk fermentations, thermophilic fermentations can be carried out at temperatures around 45°C. An important example is yogurt production. **Yogurt** is made using a special starter culture in which two major bacteria are present in a 1:1 ratio: *S. thermophilus* and *L. bulgaricus*. With these organisms growing in concert, acid is produced by *Streptococcus*, and aroma components are formed by the *Lactobacillus*. Freshly prepared yogurt contains 10⁹ bacteria per gram.

Therapeutic

Fermented milks may have beneficial therapeutic effects. Acidophilus milk is produced by using *Lactobacillus acidophilus*.

Table 41.7 Major Categories and Examples of Fermented Milk Products

Category	Typical Examples
I. Lactic fermentations	
Mesophilic	Buttermilk Cultured buttermilk Långofil Tötmjolk Ymer
Thermophilic	Yogurt, laban, zabadi, labneh, skyr Bulgarian buttermilk
Therapeutic	Biogarde®, Bifighurt® Acidophilus milk, yakult Cultura-AB®
II. Yeast-lactic fermentations	Kefir, koumiss, acidophilus-yeast milk
III. Mold-lactic fermentations	Viili

Source: Table 3.1, p. 58. In B. A. Law, editor. 1997. *Microbiology and Biochemistry of cheese and fermented milk*, 2nd ed. New York: Chapman and Hall.

Box 41.3

Starter Cultures, Bacteriophage Infections, and Plasmids

Cultures of lactic acid bacteria, called **starter cultures**, are added to milk during the preparation of many dairy products. For example, *Streptococcus lactis* and *S. cremoris* are used in the production of cheese. One of the greatest problems for the dairy industry is the presence of bacteriophages that destroy these starter cultures. Lactic acid production by a heavily phage-infected starter culture can come to a halt within 30 minutes. The industry has tried to overcome this problem by practicing aseptic techniques in order to reduce phage contamination, and by selecting for phage-resistant bacterial cultures.

Most efforts at control have not been successful in the longer term. It has been found that the very aseptic techniques and phage-resistant pure cultures used to attempt to solve this problem actually were parts of

the problem. The most stable and dependable cultures, called P starter cultures, contain the bacteriophages in a pseudolysogenic state. When cultures are grown without phages or under aseptic conditions (L starters), they lose their phage resistance. The key to this riddle appears to be plasmids, which can block phage adsorption. The loss of the plasmids in a subpopulation of the bacteria allows the phage carrier state to be established. New phages can develop by acquiring restriction enzymes (*see section 14.1*) from plasmids. These modified phages again become lytic, establishing a new equilibrium in the population.

Other control approaches are being tested. Antisense RNA is now being used in an attempt to provide an agent against bacteriophage genes to help in the constant struggle between lactic acid bacteria and their phages.

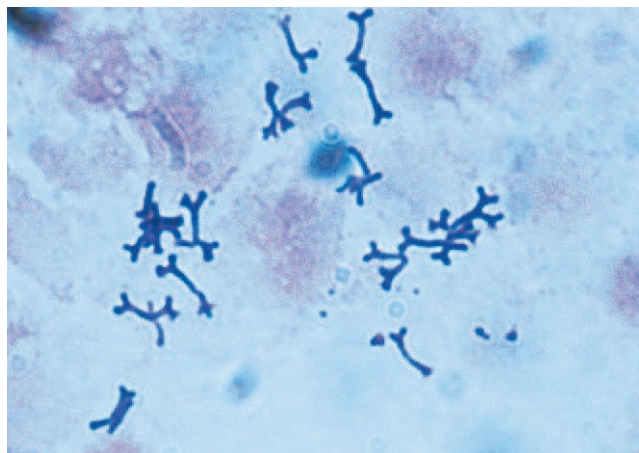


Figure 41.14 Bifidobacteria. Cultured milks are increasing in popularity. A light micrograph of *Bifidobacterium*, a microorganism suggested to provide many health benefits.

L. acidophilus may modify the microbial flora in the lower intestine, thus improving general health, and it often is used as a dietary adjunct. Many microorganisms in fermented dairy products stabilize the bowel microflora, and some appear to have antimicrobial properties. The exact nature and extent of health benefits are still unclear but may involve minimizing lactose intolerance, lowering serum cholesterol, and possibly exhibiting anticancer activity. Several lactobacilli have antitumor compounds in their cell walls. Such findings suggest that diets including lactic acid bacteria, especially *L. acidophilus*, may contribute to the control of colon cancer.

Another interesting group used in milk fermentations are the bifidobacteria. The genus *Bifidobacterium* contains irregular, nonsporing, gram-positive rods that may be club-shaped or forked at the end (figure 41.14). Bifidobacteria are nonmotile, anaerobic, and ferment lactose and other sugars to acetic and lactic acids. They are typical residents of the human intestinal tract and were discovered in 1906. Many beneficial properties are attributed to them. Bifidobacteria are thought to help maintain the normal intestinal balance, while improving lactose tolerance; to possess antitumorigenic activity; and to reduce serum cholesterol levels. In addition, some believe that they promote calcium absorption and the synthesis of B-complex vitamins. It has also been suggested that bifidobacteria reduce or prevent the excretion of rotaviruses, a cause of diarrhea among children. Bifid-amended fermented milk products are now available in various parts of the world (figure 41.15).

Yeast Lactic

Yeast-lactic fermentations include **kefir**, a product with an ethanol concentration of up to 2%. This unique fermented milk originated in the Caucasus Mountains and it is produced east into Mongolia. Kefir products tend to be foamy and frothy, due to active carbon



Figure 41.15 Examples of Bifid-Amended Dairy Products. These are produced in many countries.

dioxide production. This process is based on the use of kefir “grains” as an inoculum. These are coagulated lumps of casein that contain yeasts, lactic acid bacteria, and acetic acid bacteria (figure 41.16). In this fermentation, the grains are used to inoculate the fresh milk and then recovered at the end of the fermentation. Originally, kefir was produced in leather sacks hung by the front door during the day, and passersby were expected to push and knead the sack to mix and stimulate the fermentation. Fresh milk could be added occasionally to maintain activity.

Mold Lactic

Mold-lactic fermentation results in a unique Finnish fermented milk called viili. The milk is placed in a cup and inoculated with a mixture of the fungus *Geotrichium candidum* and lactic acid bacteria. The cream rises to the surface, and after incubation at 18 to 20°C for 24 hours, lactic acid reaches a concentration of 0.9%. The fungus forms a velvety layer across the top of the final product, which also can be made with a bottom fruit layer to add extra flavor.

1. What are the major types of milk fermentations?
2. Briefly describe how buttermilk, sour cream, and yogurt are made.
3. What is unique about the morphology of *Bifidobacterium*? Why is it used in milk?
4. How and where are kefir and viili made?

Cheese Production

Cheese is one of the oldest human foods and is thought to have been developed approximately 8,000 years ago. About 2,000 distinct varieties of cheese are produced throughout the world,

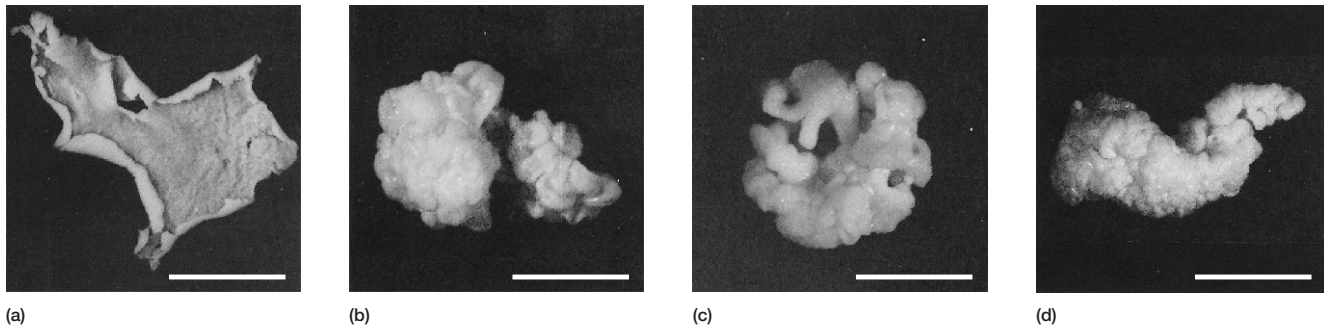


Figure 41.16 Kefir Grains Used to Inoculate Milk for a Yeast-Lactic Fermentation. The kefir grains are dried milk solids that contain a variety of desired bacteria and yeasts for production of kefir. After the fermentation is completed, the grains are recovered, washed, and used to inoculate another batch of fresh milk. Bars = 1 cm.

Table 41.8 Major Types of Cheese and Microorganisms Used in Their Production

Cheese (Country of Origin)	Contributing Microorganisms ^a	
	Earlier Stages of Production	Later Stages of Production
Soft, unripened		
Cottage	<i>Lactococcus lactis</i>	<i>Leuconostoc cremoris</i>
Cream	<i>L. cremoris</i> , <i>L. diacetylactis</i> , <i>S. thermophilus</i> , <i>L. bulgaricus</i>	
Mozzarella (Italy)	<i>S. thermophilus</i> , <i>L. bulgaricus</i>	
Soft, ripened		
Brie (France)	<i>Lactococcus lactis</i> , <i>L. cremoris</i>	<i>Penicillium camemberti</i> , <i>P. candidum</i> , <i>Brevibacterium linens</i>
Camembert (France)	<i>L. lactis</i> , <i>L. cremoris</i>	<i>Penicillium camemberti</i> , <i>Brevibacterium linens</i>
Semisoft		
Blue, Roquefort (France)	<i>Lactococcus lactis</i> , <i>L. cremoris</i>	<i>Penicillium roqueforti</i>
Brick, Muenster (United States)	<i>L. lactis</i> , <i>L. cremoris</i>	<i>Brevibacterium linens</i>
Limburger (Belgium)	<i>L. lactis</i> , <i>L. cremoris</i>	<i>Brevibacterium linens</i>
Hard, ripened		
Cheddar, Colby (Britain)	<i>Lactococcus lactis</i> , <i>L. cremoris</i>	<i>Lactobacillus casei</i> , <i>L. plantarum</i>
Swiss (Switzerland)	<i>L. lactis</i> , <i>L. helveticus</i> , <i>S. thermophilus</i>	<i>Propionibacterium shermanii</i> , <i>P. freudenreichii</i>
Very hard, ripened		
Parmesan (Italy)	<i>Lactococcus lactis</i> , <i>L. cremoris</i> , <i>S. thermophilus</i>	<i>Lactobacillus bulgaricus</i>

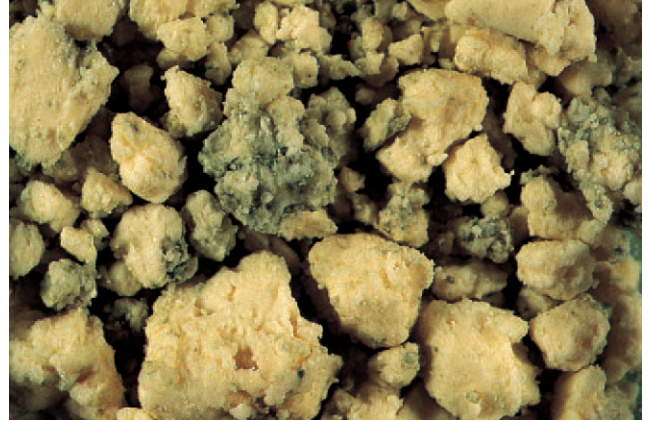
^a*Lactococcus lactis* stands for *L. lactis* subsp. *lactis*. *Lactococcus cremoris* is *L. lactis* subsp. *cremoris*, and *Lactococcus diacetylactis* is *L. lactis* subsp. *diacetylactis*.

representing approximately 20 general types (table 41.8 and figure 41.17). Often cheeses are classified based on texture or hardness as soft cheeses (cottage, cream, Brie), semisoft cheeses (Muenster, Limburger, blue), hard cheeses (cheddar, Colby, Swiss), or very hard cheeses (Parmesan). All cheese re-

sults from a lactic acid fermentation of milk, which results in coagulation of milk proteins and formation of a curd. Rennin, an enzyme from calf stomachs, but now produced by genetically engineered microorganisms, can also be used to promote curd formation. After the curd is formed, it is heated and pressed to



(a)



(b)



(c)



(d)

Figure 41.17 Cheese. A bewildering array of cheeses are produced around the world using microorganisms. (a) Gouda (top left) and cheddar cheese (lower right). Note the typical indentations on the surface of Gouda caused by the cheesecloth and red wax covering. (b) Roquefort cheese crumbled for use in salad dressing. The dark areas are the result of extensive *Penicillium* growth. (c) Swiss cheese. This hard, ripened cheese contains holes formed by carbon dioxide from a *Propionibacterium* fermentation. (d) Brie (left) and Limburger (right) cheeses are soft, ripened cheeses. Ripening results from the surface growth of microorganisms like *Penicillium camemberti* (Brie) and *Brevibacterium linens* (Limburger). (e) Cottage cheese and cream cheese (spread on crackers) are soft, unripened cheeses. They are sold immediately after production, and the curd is consumed without further modification by microorganisms.



(e)



Figure 41.18 Cheddar Cheese Production. Cheddar, a village in England, has given its name to a cheese made in many parts of the world. “Cheddaring” is the process of turning and piling the curd to express whey and develop desired cheese texture.

remove the watery part of the milk or whey, salted, and then usually ripened (**figure 41.18**). The cheese curd can be packaged for ripening with or without additional microorganisms. Cheese curd inoculation is used in the manufacture of Roquefort and blue cheese. In this case *Penicillium roqueforti* spores are added to the curds just before the final cheese processing. Sometimes the surface of an already formed cheese is inoculated at the start of ripening; for example, Camembert cheese is inoculated with spores of *Penicillium camemberti*. The final hardness of the cheese is partially a function of the length of ripening. Soft cheeses are ripened for only about 1 to 5 months, whereas hard cheeses need 3 to 12 months, and very hard cheeses like Parmesan require 12 to 16 months ripening.

The ripening process also is critical for Swiss cheese. Gas production by *Propionibacterium* contributes to final flavor development and hole or eye formation in this cheese. Some cheeses are soaked in brine to stimulate the development of specific fungi and bacteria; Limburger is one such cheese.

Meat and Fish

Besides the fermentation of dairy products, a variety of meat products, especially sausage, can be fermented: country-cured hams, summer sausage, salami, cervelat, Lebanon bologna, fish sauces (processed by halophilic *Bacillus* species), izushi, and katsubushi. *Pediococcus cerevisiae* and *Lactobacillus plantarum* are most often involved in sausage fermentations. Izushi is based on the fermentation of fresh fish, rice, and vegetables by *Lactobacillus* spp.; katsubushi results from the fermentation of tuna by *Aspergillus glaucus*. Both meat fermentations originated in Japan.

1. What major steps are used to produce cheese? How is the cheese curd formed in this process? What is whey? How does Swiss cheese get its holes?
2. Which fungal genus is often used in cheese making? What cheeses are produced using this genus?
3. Give a microbial genus used in meat fermentations.

Production of Alcoholic Beverages

A variety of plants that contain adequate carbohydrates can be used to produce alcoholic beverages. When carbohydrates are available in readily fermentable form, the fermentation can be started immediately. For example, grapes are crushed to release the juice or **must**, which can be allowed to ferment without further delay. The must also can be sterilized by pasteurization or the use of sulfur dioxide, and then the desired microbial culture added.

In contrast, before cereals and other starchy materials can be used as substrates for the production of alcohol, their complex carbohydrates must be hydrolyzed. They are mixed with water and incubated in a process called **mashing**. The insoluble material is then removed to yield the **wort**, a clear liquid containing fermentable sugars and other simple molecules. Much of the art of beer and ale production involves the controlled hydrolysis of protein and carbohydrates to provide the desired body and flavor of the final product.

Wines and Champagnes

Wine production, or the science of **enology** [Greek *oinos*, wine, and *ology*, the science of], starts with the collection of grapes, continues with their crushing and the separation of the liquid (must) before fermentation, and concludes with a variety of storage and aging steps (**figure 41.19**). All grapes have white juices. To make a red wine from a red grape, the grape skins are allowed to remain in contact with the must before fermentation to release their skin-coloring components. Wines can be created by using the natural grape skin microorganisms. This natural mixture of bacteria and yeasts gives unpredictable fermentation results. To avoid such problems, one can treat the fresh must with a sulfur dioxide fumigant and add a desired strain of *Saccharomyces cerevisiae* or *S. ellipsoideus*. After inoculation the juice is fermented for 3 to 5 days at temperatures varying between 20 and 28°C. Depending on the alcohol tolerance of the yeast strain, the final product may contain 10 to 18% alcohol. Clearing and development of flavor occur during the aging process. [Yeasts \(pp. 554, 560–63\)](#)

A critical part of wine making involves the choice of whether to produce a dry (no remaining free sugar) or a sweeter (varying amounts of free sugar) wine. This can be controlled by regulating the initial must sugar concentration. With higher levels of sugar, alcohol will accumulate and inhibit the fermentation before the sugar can be completely used, thus producing a sweeter wine. During final fermentation in the aging process, flavoring compounds accumulate and influence the bouquet of the wine.

Microbial growth during the fermentation process produces sediments, which are removed during **racking**. Racking can be carried out at the time the fermented wine is transferred to bottles or casks for aging or even after the wine is placed in bottles.